

Chapter 26 Check

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26 Supersymmetric Field Theories

Now we know the structure of the most general supersymmetry algebras, and we have seen how to work out the implications of this symmetry for the particle spectrum. In order to learn what supersymmetry has to say about particle interactions, we need to see how to construct supersymmetric field theories.

Originally the construction of field supermultiplets was done directly, by a repeated use of the Jacobi identities, much as in the construction of supermultiplets of one-particle states in Sections 25.4 and 25.5. Section 26.1 presents one example of this technique, used here to construct supermultiplets containing only scalar and Dirac fields. Fortunately there is an easier technique, invented by Salam and Strathdee [1], in which supermultiplets of fields are gathered into ‘superfields,’ which depend on fermionic coordinates as well as on the usual four coordinates of spacetime. Superfields are introduced in Section 26.2, and used to construct supersymmetric field theories and to study some of their consequences in Sections 26.3–26.8. This chapter will be concerned only with $N = 1$ supersymmetry, where the superfield formalism has been chiefly useful. At the end of the next chapter we will construct theories with N -extended supersymmetry by imposing the $U(N)$ R -symmetry on theories of $N = 1$ superfields.

26.1 Direct Construction of Field Supermultiplets

To illustrate the direct construction of a field supermultiplet, we will consider fields that can destroy the particles belonging to the simplest supermultiplet of arbitrary mass discussed in Section 25.5: two spinless particles and one particle of spin $1/2$. We saw in Eq. (??) that the one-particle zero spin state $|0,0\rangle$ is annihilated by the supersymmetry generator Q_a

but not by Q_a^* , so we would expect the scalar field $\phi(x)$ that creates this particle from the vacuum (which is assumed to be annihilated by all supersymmetry generators) to commute with Q_a but not with Q_a^* . That is,

$$[Q_a, \phi(x)] = 0, \quad (26.1.1)$$

$$-i \sum_b e_{ab} [Q_b^*, \phi(x)] \equiv \zeta_a(x) \neq 0. \quad (26.1.2)$$

The antisymmetric $2 \cdot 2$ matrix e_{ab} (with $e_{1/2, -1/2} \equiv +1$) is introduced here because it is $\sum_b e_{ab} Q_b^*$ that transforms under the homogeneous Lorentz group according to the $(1/2, 0)$ representation. It follows that $\zeta_a(x)$ is a two-component spinor field that also belongs to the $(1/2, 0)$ representation of the homogeneous Lorentz group.¹

From Eqs. (26.1.1)–(26.1.2) and the anticommutation relation (??), we find

$$\{Q_b, \zeta_a\} = -i \sum_c e_{ac} [\{Q_b, Q_c^*\}, \phi(x)] = 2i(\sigma^\mu e)_{ba} [P_\mu, \phi],$$

and so

$$\{Q_b, \zeta_a(x)\} = -2(\sigma^\mu e)_{ba} \partial_\mu \phi(x). \quad (26.1.3)$$

On the other hand, Eq. (26.1.2) and the anticommutation relation (??) gives

$$-i \sum_c e_{ac} \{Q_b^*, \zeta_c\} = \{Q_b^*, [Q_a^*, \phi]\} = -\{Q_a^*, [Q_b^*, \phi]\} = i \sum_c e_{bc} \{Q_a^*, \zeta_c\},$$

so $\sum_c e_{ac} \{Q_b^*, \zeta_c\}$ is antisymmetric, and therefore proportional to the antisymmetric $2 \cdot 2$ matrix e_{ab} :

$$i\{Q_b^*, \zeta_a(x)\} = 2\delta_{ab} \mathcal{F}(x). \quad (26.1.4)$$

Lorentz invariance requires that the coefficient $\mathcal{F}(x)$ is a scalar field.

We must now go one step further, and calculate the commutators of the supersymmetry generators with $\mathcal{F}(x)$. Using Eqs. (26.1.4), (26.1.2), and (??), we have

$$\delta_{ab} [Q_c^*, \mathcal{F}] = \frac{1}{2} i [Q_c^*, \{Q_b^*, \zeta_a\}] = \frac{1}{2} i [\{Q_c^*, \zeta_a\}, Q_b^*] = -\delta_{ac} [Q_b^*, \mathcal{F}].$$

By taking $a = b \neq c$, we find that this commutator vanishes:

$$[Q_c^*, \mathcal{F}(x)] = 0. \quad (26.1.5)$$

Finally, using Eqs. (26.1.4), (??), and (26.1.3), we have

$$\begin{aligned} \delta_{ab} [Q_c, \mathcal{F}] &= \frac{1}{2} i [Q_c, \{Q_b^*, \zeta_a\}] = \frac{1}{2} i [\{Q_c, Q_b^*\}, \zeta_a] - \frac{1}{2} i [Q_b^*, \{Q_c, \zeta_a\}] \\ &= -\sigma_{cb}^\mu \partial_\mu \zeta_a + i(\sigma^\mu e)_{ca} [Q_b^*, \partial_\mu \phi] \\ &= -\sigma_{cb}^\mu \partial_\mu \zeta_a + \sum_d e_{bd} (\sigma^\mu e)_{ca} \partial_\mu \zeta_d. \end{aligned}$$

¹At this point we are not assuming anything about the masses or interactions of the particles described by these fields, but it may be noted that, as explained in Section 5.9, a $(1/2, 0)$ free field can create massless particles only of helicity $+1/2$, in agreement with the result (??) that the massless spinless one-particle state $|0, 0\rangle$ that is annihilated by Q_a is in a supermultiplet with a state of helicity $+1/2$.

Contracting with δ_{ab} , this becomes

$$[Q_c, \mathcal{F}(x)] = - \sum_a \sigma_{ca}^\mu \partial_\mu \zeta_a(x). \quad (26.1.6)$$

Eqs. (26.1.1)–(26.1.6) show that the fields $\phi(x)$, $\zeta_a(x)$, and $\mathcal{F}(x)$ furnish a complete representation of the supersymmetry algebra. These fields are not Hermitian, so their complex conjugates furnish another supermultiplet:

$$[Q_a^*, \phi^*(x)] = 0, \quad (26.1.7)$$

$$-i \sum_b e_{ab} [Q_b, \phi^*(x)] = \zeta_a^*(x), \quad (26.1.8)$$

$$\{Q_b^*, \zeta_a^*(x)\} = 2(e\sigma^\mu)_{ab} \partial_\mu \phi^*(x), \quad (26.1.9)$$

$$-i\{Q_b, \zeta_a^*(x)\} = 2\delta_{ab} \mathcal{F}^*(x), \quad (26.1.10)$$

$$[Q_c, \mathcal{F}^*(x)] = 0, \quad (26.1.11)$$

$$[Q_c^*, \mathcal{F}^*(x)] = \sum_a \sigma_{ac}^\mu \partial_\mu \zeta_a^*(x). \quad (26.1.12)$$

We can express these commutation and anticommutation relations as transformation rules under a supersymmetry transformation, which shifts any bosonic or fermionic field operator $\mathcal{O}(x)$ by the infinitesimal amount

$$\delta \mathcal{O}(x) \equiv \left[\sum_a (\epsilon_a^* Q_a + \epsilon_a Q_a^*), \mathcal{O}(x) \right], \quad (26.1.13)$$

where ϵ_a is an infinitesimal fermionic c-number spinor. (Because ϵ_a and ϵ_a^* anticommute with Q_a and Q_a^* , the quantity $\epsilon_a^* Q_a + \epsilon_a Q_a^*$ is anti-Hermitian, so Eq. (26.1.13) gives $(\delta \mathcal{O})^\dagger = \delta \mathcal{O}^\dagger$.) The commutation and anticommutation rules (26.1.1)–(26.1.6) are equivalent to the transformation rules

$$\delta \phi(x) = -i \sum_{ab} \epsilon_a e_{ab} \zeta_b(x), \quad (26.1.14)$$

$$\delta \zeta_a(x) = -2 \sum_b \epsilon_b^* (\sigma^\mu e)_{ba} \partial_\mu \phi(x) - 2i \epsilon_a \mathcal{F}(x), \quad (26.1.15)$$

$$\delta \mathcal{F}(x) = - \sum_{ab} \epsilon_b^* \sigma_{ba}^\mu \partial_\mu \zeta_a(x). \quad (26.1.16)$$

This may be put into a four-component Dirac notation by introducing an infinitesimal, Majorana², four-component, spinor transformation parameter

$$\alpha \equiv -i \left(\sum_b \epsilon_a e_{ab} \epsilon_b^* \right), \quad (26.1.17)$$

so that Eq. (26.1.13) becomes

$$i\delta\mathcal{O}(x) \equiv [\bar{\alpha}Q, \mathcal{O}(x)]. \quad (26.1.18)$$

The transformation rules (26.1.14)–(26.1.16) and their complex conjugates may be put into a convenient covariant form by introducing a set of real bosonic fields A , B , F , and G , defined by

$$\frac{A + iB}{\sqrt{2}} \equiv \phi, \quad \frac{F - iG}{\sqrt{2}} \equiv \mathcal{F}, \quad (26.1.19)$$

and a four-component Majorana spinor ψ , defined by

$$\psi \equiv \frac{1}{\sqrt{2}} \left(- \sum_b \zeta_a e_{ab} \zeta_b^* \right). \quad (26.1.20)$$

Let's also recall the relation between the $4 \cdot 4$ Dirac matrices and the $2 \cdot 2$ matrices σ_μ :

$$\gamma_\mu = \begin{pmatrix} 0 & -ie\sigma_\mu^T e \\ i\sigma_\mu & 0 \end{pmatrix}.$$

The transformation rules now take the form

$$\begin{aligned} \delta A &= \bar{\alpha}\psi, & \delta B &= -i\bar{\alpha}\gamma_5\psi, \\ \delta\psi &= \partial_\mu(A + i\gamma_5 B)\gamma^\mu\alpha + (F - i\gamma_5 G)\alpha, & & \\ \delta F &= \bar{\alpha}\gamma^\mu\partial_\mu\psi, & \delta G &= -i\bar{\alpha}\gamma_5\gamma^\mu\partial_\mu\psi. \end{aligned} \quad (26.1.21)$$

A direct but tedious calculation shows that this transformation leaves invariant the action

$$\begin{aligned} I = \int d^4x \Big\{ & -\frac{1}{2}\partial_\mu A \partial^\mu A - \frac{1}{2}\partial_\mu B \partial^\mu B - \frac{1}{2}\bar{\psi}\gamma^\mu\partial_\mu\psi \\ & + \frac{1}{2}(F^2 + G^2) + m \left[FA + GB - \frac{1}{2}\bar{\psi}\psi \right] \\ & + g \left[F(A^2 + B^2) + 2GAB - \bar{\psi}(A + i\gamma_5 B)\psi \right] \Big\}. \end{aligned} \quad (26.1.22)$$

²With the phase convention we will be using here, a Majorana four-component spinor is formed from a $(1/2, 0)$ two-component spinor u_a as

$$\begin{pmatrix} u \\ -eu^* \end{pmatrix}.$$

Eq. (26.1.17) fits this definition, with $u = -i\epsilon$. Equivalently, a Majorana spinor can be formed from a two-component $(0, 1/2)$ spinor v_a as

$$\begin{pmatrix} ev^* \\ v \end{pmatrix}.$$

An example is provided by Eq. (??). Properties of Majorana spinors are considered in detail in the appendix to this chapter.

Eqs. (26.1.21) and (26.1.22) agree with the transformation rules (??) and Lagrangian density (??) found in the original work of Wess and Zumino. In the next three sections we will explore a convenient technique for checking the supersymmetry of Eq. (26.1.22) and for deriving more general supersymmetric actions.

Where the fermion field $\psi(x)$ satisfies the free-field Dirac equation $(\gamma^\mu \partial_\mu + m)\psi = 0$, these transformation rules show that $F + mA$ and $G + mB$ are invariant, and therefore commute with Q_a and Q_a^* , and hence also with P_μ . This does not prove that $F = -mA$ and $G = -mB$, but without changing any of the commutation and anticommutation rules (26.1.1)–(26.1.6) or transformation rules (26.1.21), we can redefine the fields F and G by subtracting the constants $F + mA$ and $B + mG$, respectively, so that the new fields F and G are given by $F = -mA$ and $G = -mB$, and therefore $\mathcal{F} = -m\phi^*$. This is not true in the presence of interactions, but even in the interacting case $\mathcal{F}(x)$, $F(x)$, and $G(x)$ are typically auxiliary fields that can be expressed in terms of the other fields of the supermultiplet, as is the case for the action (26.1.22).

26.2 General Superfields

It is straightforward to construct supermultiplets of fields by the direct technique illustrated in the previous section, but in order to construct supersymmetric actions we also need to know how to multiply field supermultiplets to make other supermultiplets. A great deal of work can be saved by using a formalism invented by Salam and Strathdee [1], in which the fields in any supermultiplet are assembled into a single superfield.

Just as the four-momentum operators P_μ are defined as the generators of translations of the ordinary spacetime coordinates x^μ , the four supersymmetry generators Q_a and $Q_a^\dagger$ may be regarded as the generators of translations of four fermionic c-number superspace coordinates, which anticommute with each other and with fermionic fields but commute with the x^μ and all bosonic fields. We aim at constructing Lorentz-invariant Lagrangian densities, so it will be convenient to adopt the four-component Dirac formalism described in Section 25.2. The supersymmetry generators are gathered into a four-component Majorana spinor Q_α , and correspondingly, the superspace coordinates are gathered into another four-component Majorana spinor θ_α . (Various properties of Majorana spinors are outlined in the appendix to this chapter.) The supersymmetry generators have non-vanishing anticommutators, so we cannot take them as simply proportional to the supercoordinate translation operators $\partial/\partial\theta_\alpha$. Instead, Salam and Strathdee found that the supersymmetry algebra would be satisfied if we suppose that the commutator or anticommutator of the supersymmetry generator Q with any bosonic or fermionic superfield $S(x, \theta)$ is

$$[Q, S] = i\mathcal{Q}S, \quad (26.2.1)$$

where $\mathcal{Q}$ is the superspace differential operator

$$\mathcal{Q} \equiv -\frac{\partial}{\partial\bar{\theta}} + \gamma^\mu \theta \frac{\partial}{\partial x^\mu}. \quad (26.2.2)$$

(As usual $\bar{\theta} \equiv \theta^\dagger \beta$. All derivatives with respect to fermionic c-number variables should be understood as *left*-derivatives, calculated by moving the variable to the left of any expression

before differentiating with respect to it.) For Majorana spinors $\bar{\theta} = \theta^T \gamma_5 \epsilon$, with the $4 \cdot 4$ matrix ϵ given by Eq. (26.A.3), so Eq. (26.2.1) may be expressed more explicitly as

$$\mathcal{Q}_\alpha = \sum_\gamma (\gamma_5 \epsilon)_{\alpha\gamma} \frac{\partial}{\partial \theta_\gamma} + \sum_\gamma \gamma_{\alpha\gamma}^\mu \theta_\gamma \frac{\partial}{\partial x^\mu}. \quad (26.2.3)$$

Likewise,

$$\bar{\mathcal{Q}}_\beta = \sum_\gamma \mathcal{Q}_\gamma (\gamma_5 \epsilon)_{\gamma\beta} = \frac{\partial}{\partial \theta_\beta} - \sum_\gamma (\gamma_5 \epsilon \gamma^\mu)_{\beta\gamma} \theta_\gamma \frac{\partial}{\partial x^\mu}. \quad (26.2.4)$$

It is straightforward to calculate that

$$\{\mathcal{Q}_\alpha, \bar{\mathcal{Q}}_\beta\} = (\gamma_5 \epsilon \gamma^\mu \gamma_5 \epsilon)_{\beta\alpha} \frac{\partial}{\partial x^\mu} + \gamma_{\alpha\beta}^\mu \frac{\partial}{\partial x^\mu}. \quad (26.2.5)$$

But Eq. (??) shows that $\gamma_\mu^T = -\mathcal{C} \gamma_\mu \mathcal{C}^{-1}$, where $\mathcal{C}$ is the matrix $\mathcal{C} = -\gamma_5 \epsilon$, so both terms on the right-hand side of Eq. (26.2.5) are equal, and therefore

$$\{\mathcal{Q}_\alpha, \bar{\mathcal{Q}}_\beta\} = 2\gamma_{\alpha\beta}^\mu \frac{\partial}{\partial x^\mu}. \quad (26.2.6)$$

Eqs. (26.2.6) and (26.2.1) together with the generalized Jacobi identities (??) show that

$$[\{Q_\alpha, \bar{Q}_\beta\}, S] = \{\mathcal{Q}_\alpha, \bar{\mathcal{Q}}_\beta\} S = 2\gamma_{\alpha\beta}^\mu \partial_\mu S = -2i\gamma_{\alpha\beta}^\mu [P_\mu, S], \quad (26.2.7)$$

in agreement with the anticommutation relation (??).

It is often more convenient to express the commutation and anticommutation relations (26.2.1) as transformation rules under infinitesimal supersymmetry transformations. Combining Eqs. (26.1.18), (26.2.1), and (26.2.2) shows that a supersymmetry transformation with infinitesimal Majorana spinor parameter α changes a superfield $S(x, \theta)$ by an amount

$$\delta S = (\bar{\alpha} \mathcal{Q}) S = -\left(\bar{\alpha} \frac{\partial S}{\partial \theta}\right) + (\bar{\alpha} \gamma^\mu \theta) \frac{\partial S}{\partial x^\mu}. \quad (26.2.8)$$

Recall that $\partial/\partial \bar{\theta}$ here acts on the left of any expression. In particular, where M is any linear combination of the matrices $\mathbf{1}$, $\gamma_5 \gamma_\mu$, and γ_5 for which $\bar{\theta} M \theta$ does not vanish, we have $\bar{\theta}' M \theta'' = \bar{\theta}'' M \theta'$, so

$$\frac{\partial}{\partial \bar{\theta}} (\bar{\theta} M \theta) = 2M \theta. \quad (26.2.9)$$

The components of θ anticommute, so any product of their components vanishes if two of them are the same component. But θ has only four components, so any function of θ has a power series that terminates with its quartic term. Furthermore, as shown in the appendix to this chapter, a product of two θ s is proportional to a linear combination of $(\bar{\theta}\theta)$, $(\bar{\theta}\gamma_\mu\gamma_5\theta)$, and $(\bar{\theta}\gamma_5\theta)$; a product of three θ s is proportional to $(\bar{\theta}\gamma_5\theta)\theta$; and a product of four θ s is proportional to $(\bar{\theta}\gamma_5\theta)^2$. The most general function of x^μ and θ may therefore be expressed as

$$\begin{aligned} S(x, \theta) = & C(x) - i(\bar{\theta}\gamma_5\omega(x)) - \frac{i}{2}(\bar{\theta}\gamma_5\theta)M(x) - \frac{1}{2}(\bar{\theta}\theta)N(x) \\ & + \frac{i}{2}(\bar{\theta}\gamma_5\gamma_\mu\theta)V^\mu(x) - i(\bar{\theta}\gamma_5\theta)\left(\bar{\theta}\left[\lambda(x) + \frac{1}{2}\not{\partial}\omega(x)\right]\right) \\ & - \frac{1}{4}(\bar{\theta}\gamma_5\theta)^2\left(D(x) + \frac{1}{2}\square C(x)\right). \end{aligned} \quad (26.2.10)$$

(The terms $\frac{1}{2}\not{\partial}\omega$ and $\frac{1}{2}\square C(x)$ are separated from $\lambda(x)$ and $D(x)$, respectively, for later convenience.) If $S(x, \theta)$ is a scalar then $C(x)$, $M(x)$, $N(x)$, and $D(x)$ are scalar (or pseudoscalar) fields; $\omega(x)$ and $\lambda(x)$ are four-component spinor fields; and $V^\mu(x)$ is a vector field. Also, using the reality properties of bilinear products of Majorana fields given in the appendix to this chapter, we see that if $S(x, \theta)$ is real, then $C(x)$, $M(x)$, $N(x)$, $V^\mu(x)$, and $D(x)$ are all real, while $\omega(x)$ and $\lambda(x)$ are Majorana spinors satisfying the phase convention $s^* = -\beta\gamma_5\epsilon s$.

Now we must work out the supersymmetry transformation properties of the component fields in Eq. (26.2.10). Applying Eqs. (26.2.8) and (26.2.9) to the expansion (26.2.10) gives

$$\begin{aligned}\delta S = & (\bar{\alpha}\gamma^\mu\theta)\frac{\partial C}{\partial x^\mu} + i(\bar{\alpha}\gamma_5\omega) - i(\bar{\alpha}\gamma^\mu\theta)\left(\bar{\theta}\gamma_5\frac{\partial\omega}{\partial x^\mu}\right) \\ & + i(\bar{\alpha}\gamma_5\theta)M - \frac{i}{2}(\bar{\alpha}\gamma^\mu\theta)(\bar{\theta}\gamma_5\theta)\frac{\partial M}{\partial x^\mu} \\ & + (\bar{\alpha}\theta)N - \frac{1}{2}(\bar{\alpha}\gamma^\mu\theta)(\bar{\theta}\theta)\frac{\partial N}{\partial x^\mu} \\ & - i(\bar{\alpha}\gamma_5\gamma_\nu\theta)V^\nu + \frac{i}{2}(\bar{\alpha}\gamma^\mu\theta)(\bar{\theta}\gamma_5\gamma_\nu\theta)\frac{\partial V^\nu}{\partial x^\mu} \\ & + 2i(\bar{\alpha}\gamma_5\theta)\left(\bar{\theta}\left[\lambda + \frac{1}{2}\not{\partial}\omega\right]\right) + i(\bar{\theta}\gamma_5\theta)\left(\bar{\alpha}\left[\lambda + \frac{1}{2}\not{\partial}\omega\right]\right) \\ & - i(\bar{\alpha}\gamma^\mu\theta)(\bar{\theta}\gamma_5\theta)\left(\bar{\theta}\partial_\mu\left[\lambda + \frac{1}{2}\not{\partial}\omega\right]\right) \\ & + (\bar{\theta}\gamma_5\theta)(\bar{\alpha}\gamma_5\theta)\left[D + \frac{1}{2}\square C\right].\end{aligned}$$

We need to put each term in the standard form of Eq. (26.2.10). For this purpose, we note that: Eq. (26.A.9) gives

$$\begin{aligned}(\bar{\alpha}\gamma^\mu\theta)(\bar{\theta}\gamma_5\partial_\mu\omega) = & -\frac{1}{4}(\bar{\theta}\theta)(\bar{\alpha}\not{\partial}\gamma_5\omega) - \frac{1}{4}(\bar{\theta}\gamma_5\gamma^\nu\theta)(\bar{\alpha}\not{\partial}\gamma_\nu\omega) \\ & - \frac{1}{4}(\bar{\theta}\gamma_5\theta)(\bar{\alpha}\not{\partial}\omega);\end{aligned}$$

Eq. (26.A.16) gives

$$(\bar{\alpha}\gamma^\mu\theta)(\bar{\theta}\theta) = -(\bar{\alpha}\gamma^\mu\gamma_5\theta)(\bar{\theta}\gamma_5\theta);$$

Eq. (26.A.17) gives

$$(\bar{\alpha}\gamma^\mu\theta)(\bar{\theta}\gamma_5\gamma_\nu\theta) = -(\bar{\alpha}\gamma^\mu\gamma_\nu\theta)(\bar{\theta}\gamma_5\theta);$$

Eq. (26.A.9) gives

$$\begin{aligned}(\bar{\alpha}\gamma_5\theta)\left(\bar{\theta}\left[\lambda + \frac{1}{2}\not{\partial}\omega\right]\right) \\ = & -\frac{1}{4}(\bar{\theta}\theta)\left(\bar{\alpha}\gamma_5\left[\lambda + \frac{1}{2}\not{\partial}\omega\right]\right) + \frac{1}{4}(\bar{\theta}\gamma_5\gamma^\mu\theta)\left(\bar{\alpha}\gamma_\mu\left[\lambda + \frac{1}{2}\not{\partial}\omega\right]\right) \\ & - \frac{1}{4}(\bar{\theta}\gamma_5\theta)\left(\bar{\alpha}\left[\lambda + \frac{1}{2}\not{\partial}\omega\right]\right);\end{aligned}$$

and Eq. (26.A.19) gives

$$\begin{aligned} & (\bar{\alpha}\gamma^\mu\theta)(\bar{\theta}\gamma_5\theta) \left(\bar{\theta}\partial_\mu \left[\lambda + \frac{1}{2}\not{\phi}\omega \right] \right) \\ &= -\frac{1}{4} \left(\bar{\alpha}\not{\phi}\gamma_5 \left[\lambda + \frac{1}{2}\not{\phi}\omega \right] \right) (\bar{\theta}\gamma_5\theta)^2. \end{aligned}$$

Using these relations and rearranging terms in order of increasing numbers of θ factors, we have

$$\begin{aligned} \delta S &= i(\bar{\alpha}\gamma_5\omega) + (\bar{\alpha}[\not{\phi}C + i\gamma_5M + N - i\gamma_5\dot{V}]\theta) \\ &\quad - \frac{i}{2}(\bar{\theta}\theta) (\bar{\alpha}\gamma_5[\lambda + \not{\phi}\omega]) + \frac{i}{2}(\bar{\theta}\gamma_5\theta) (\bar{\alpha}[\lambda + \not{\phi}\omega]) \\ &\quad + \frac{i}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta)(\bar{\alpha}\gamma_\mu\lambda) + \frac{i}{2}(\bar{\theta}\gamma_5\gamma^\nu\theta)(\bar{\alpha}\partial_\nu\omega) \\ &\quad + \frac{1}{2}(\bar{\theta}\gamma_5\theta) (\bar{\alpha}[-i\not{\phi}M - \gamma_5\not{\phi}N - i\not{\phi}\dot{V} + \gamma_5(D + \frac{1}{2}\square C)]\theta) \\ &\quad - \frac{i}{4}(\bar{\theta}\gamma_5\theta)^2 (\bar{\alpha}\gamma_5[\not{\phi}\lambda + \frac{1}{2}\square\omega]), \end{aligned}$$

or, using the symmetry property (26.A.7),

$$\begin{aligned} \delta S &= i(\bar{\alpha}\gamma_5\omega) + (\bar{\theta}[-\not{\phi}C + i\gamma_5M + N - i\gamma_5\dot{V}]\alpha) \\ &\quad - \frac{i}{2}(\bar{\theta}\theta) (\bar{\alpha}\gamma_5[\lambda + \not{\phi}\omega]) + \frac{i}{2}(\bar{\theta}\gamma_5\theta) (\bar{\alpha}[\lambda + \not{\phi}\omega]) \\ &\quad + \frac{i}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta)(\bar{\alpha}\gamma_\mu\lambda) + \frac{i}{2}(\bar{\theta}\gamma_5\gamma^\nu\theta)(\bar{\alpha}\partial_\nu\omega) \\ &\quad + \frac{1}{2}(\bar{\theta}\gamma_5\theta) (\bar{\theta}[i\not{\phi}M - \gamma_5\not{\phi}N - i\partial_\mu\dot{V}\gamma^\mu + \gamma_5(D + \frac{1}{2}\square C)]\alpha) \\ &\quad - \frac{i}{4}(\bar{\theta}\gamma_5\theta)^2 (\bar{\alpha}\gamma_5[\not{\phi}\lambda + \frac{1}{2}\square\omega]). \end{aligned}$$

If we now compare this with the terms up to second order in θ in the expansion (26.2.10), we find the transformation rules:

$$\delta C = i(\bar{\alpha}\gamma_5\omega), \quad (26.2.11)$$

$$\delta\omega = (-i\gamma_5\not{\phi}C - M + i\gamma_5N + \dot{V})\alpha, \quad (26.2.12)$$

$$\delta M = -(\bar{\alpha}[\lambda + \not{\phi}\omega]), \quad (26.2.13)$$

$$\delta N = i(\bar{\alpha}\gamma_5[\lambda + \not{\phi}\omega]), \quad (26.2.14)$$

$$\delta V_\mu = (\bar{\alpha}\gamma_\mu\lambda) + (\bar{\alpha}\partial_\mu\omega). \quad (26.2.15)$$

The terms of third and fourth order in θ give

$$\begin{aligned} \delta \left[\lambda + \frac{1}{2}\not{\phi}\omega \right] &= \frac{1}{2} \left[-\not{\phi}M - i\gamma_5\not{\phi}N + \partial_\mu\dot{V}\gamma^\mu + i\gamma_5 \left(D + \frac{1}{2}\square C \right) \right] \alpha, \\ \delta \left[D + \frac{1}{2}\square C \right] &= i \left(\bar{\alpha}\gamma_5 \left[\not{\phi}\lambda + \frac{1}{2}\square\omega \right] \right). \end{aligned}$$

Combining the latter two transformation rules with the transformation rules (26.2.11) and (26.2.12) for C and ω yields much simpler transformation rules for λ and D :

$$\delta\lambda = \left(\frac{1}{2}[\partial_\mu \not{V}, \gamma^\mu] + i\gamma_5 D \right) \alpha, \quad (26.2.16)$$

$$\delta D = i(\bar{\alpha}\gamma_5 \not{\partial}\lambda). \quad (26.2.17)$$

It is to achieve this simplification that the terms $\frac{1}{2}\not{\partial}\omega$ and $\frac{1}{2}\square C$ were separated from λ and D in the expansion (26.2.10).

The whole point of the superfield formalism is to simplify the task of making supermultiplets out of other supermultiplets. Given two superfields S_1 and S_2 that both satisfy the transformation rule (26.2.8), their product $S \equiv S_1 S_2$ satisfies

$$\begin{aligned} \delta S &= [(\bar{\alpha}Q), S_1 S_2] = (\delta S_1) S_2 + S_1 (\delta S_2) \\ &= ((\bar{\alpha}\mathcal{Q})S_1) S_2 + S_1 ((\bar{\alpha}\mathcal{Q})S_2) = (\bar{\alpha}\mathcal{Q})S, \end{aligned} \quad (26.2.18)$$

and is therefore also a superfield. A straightforward calculation using Eqs. (26.A.7), (26.A.16), (26.A.18), and (26.A.19) gives its components as

$$C = C_1 C_2, \quad (26.2.19)$$

$$\omega = C_1 \omega_2 + C_2 \omega_1, \quad (26.2.20)$$

$$M = C_1 M_2 + C_2 M_1 + \frac{i}{2}(\bar{\omega}_1 \gamma_5 \omega_2), \quad (26.2.21)$$

$$N = C_1 N_2 + C_2 N_1 - \frac{1}{2}(\bar{\omega}_1 \omega_2), \quad (26.2.22)$$

$$V^\mu = C_1 V_2^\mu + C_2 V_1^\mu - \frac{i}{2}(\bar{\omega}_1 \gamma_5 \gamma^\mu \omega_2), \quad (26.2.23)$$

$$\begin{aligned} \lambda &= C_1 \lambda_2 + C_2 \lambda_1 - \frac{1}{2}\gamma^\mu \omega_1 \partial_\mu C_2 - \frac{1}{2}\gamma^\mu \omega_2 \partial_\mu C_1 + \frac{i}{2}\not{V}_1 \gamma_5 \omega_2 + \frac{i}{2}\not{V}_2 \gamma_5 \omega_1 \\ &\quad + \frac{1}{2}(N_1 - i\gamma_5 M_1)\omega_2 + \frac{1}{2}(N_2 - i\gamma_5 M_2)\omega_1, \end{aligned} \quad (26.2.24)$$

$$\begin{aligned} D &= -\partial_\mu C_1 \partial^\mu C_2 + C_1 D_2 + C_2 D_1 + M_1 M_2 + N_1 N_2 \\ &\quad - \left(\bar{\omega}_1 \left[\lambda_2 + \frac{1}{2}\not{\partial}\omega_2 \right] \right) - \left(\bar{\omega}_2 \left[\lambda_1 + \frac{1}{2}\not{\partial}\omega_1 \right] \right) - V_{1\mu} V_2^\mu. \end{aligned} \quad (26.2.25)$$

It is trivial that linear combinations of superfields are superfields, in the same sense, and that spacetime derivatives and complex conjugates of superfields are superfields. But multiplying a superfield by some function of θ or differentiating it with respect to θ does not in general yield a superfield. (For instance, θ itself is evidently not a superfield, because θ is a fermionic c-number and therefore commutes with $\bar{\alpha}Q$, while $\mathcal{Q}\theta \neq 0$.) There is, however, a way of combining a derivative of a superfield with respect to θ and multiplying it by a factor θ which does yield another superfield.

Consider the superspace differential operator $\mathcal{D}_\alpha$ defined by

$$\mathcal{D} \equiv -\frac{\partial}{\partial \bar{\theta}} - \gamma^\mu \theta \frac{\partial}{\partial x^\mu}, \quad (26.2.26)$$

or more explicitly

$$\mathcal{D}_\alpha = \sum_\gamma (\gamma_5 \epsilon)_{\alpha\gamma} \frac{\partial}{\partial \theta_\gamma} - \sum_\gamma \gamma_{\alpha\gamma}^\mu \theta_\gamma \frac{\partial}{\partial x^\mu}. \quad (26.2.27)$$

The only difference between the definitions of $\mathcal{D}$ and $\mathcal{Q}$ is a change of sign of the terms involving the spacetime derivative. In consequence of this change of sign, in the anticommutator of $\mathcal{D}_\beta$ with $\mathcal{Q}_\alpha$, instead of getting two equal terms like those in Eq. (26.2.5) with the same sign, we get them with opposite signs, so that they cancel:

$$\{\mathcal{D}_\beta, \mathcal{Q}_\alpha\} = 0. \quad (26.2.28)$$

Since α is fermionic, it follows that $(\bar{\alpha}\mathcal{Q})$ commutes with $\mathcal{D}_\beta$, so if $S(x, \theta)$ is a superfield, then

$$\delta \mathcal{D}_\beta S \equiv -i[(\bar{\alpha}\mathcal{Q}), \mathcal{D}_\beta S] = -i\mathcal{D}_\beta[(\bar{\alpha}\mathcal{Q}), S] = \mathcal{D}_\beta(\bar{\alpha}\mathcal{Q})S = (\bar{\alpha}\mathcal{Q})\mathcal{D}_\beta S, \quad (26.2.29)$$

so that $\mathcal{D}_\beta S$ is also a superfield. *Thus an arbitrary polynomial function of superfields S and their superderivatives $\mathcal{D}_\beta S$, $\mathcal{D}_\beta \mathcal{D}_\gamma S$, etc. is also a superfield.*

It is not necessary to add here that in constructing a superfield out of other superfields we can also include their spacetime derivatives, because these can be obtained from second superderivatives. Since the only difference between $\mathcal{D}_\beta$ and $\mathcal{Q}_\beta$ is in the sign of the term involving ∂_μ , the anticommutators of the $\mathcal{D}$ s are the same as those of the $\mathcal{Q}$ s, except for a change of sign:

$$\{\mathcal{D}_\alpha, \bar{\mathcal{D}}_\beta\} = -2\gamma_{\alpha\beta}^\mu \frac{\partial}{\partial x^\mu}. \quad (26.2.30)$$

Now let us consider how to construct a supersymmetric action out of superfields. There is no such thing as a supersymmetric Lagrangian density, because the anticommutation relation (26.2.6) shows that if $\delta\mathcal{L} = 0$ then $\mathcal{L}$ must be a constant. Even if the Lagrangian density is not supersymmetric, the action will still be supersymmetric if $\delta\mathcal{L}(x)$ is a derivative, which would not contribute to $\delta \int \mathcal{L} d^4x$. In general, the Lagrangian density $\mathcal{L}$ can be written as a sum of terms, each of which is some component of a superfield that is constructed out of elementary superfields and their superderivatives. Inspection of the transformation rules (26.2.11)–(26.2.17) for the individual components shows that in the absence of any special conditions on a general superfield, the only component of such a superfield whose variation is a derivative is the D -component. Also, for the D -component of any superfield to be a scalar, the superfield itself must be a scalar. Therefore, unless there are special conditions on the individual superfields from which the Lagrangian density is constructed, a supersymmetric action can only be the integral of the D -term of a scalar superfield Λ :

$$I = \int d^4x [\Lambda]_D. \quad (26.2.31)$$

But in fact no action of this sort would be physically satisfactory without special conditions on the superfields from which it is constructed. For a general superfield $S(x, \theta)$, the only sort of supersymmetric kinematic action I_0 that is bilinear in S and S^* and involves no more than two derivatives of the component fields is of the form

$$I_0 \propto \int d^4x [S^* S]_D. \quad (26.2.32)$$

From Eq. (26.2.25), we see that S^*S has the D -component

$$\begin{aligned}
[S^*S]_D = & -\partial_\mu C^* \partial^\mu C - \frac{1}{2}(\bar{\omega} \gamma^\mu \partial_\mu \omega) + \frac{1}{2}((\partial_\mu \bar{\omega}) \gamma^\mu \omega) \\
& + C^* D + D^* C - (\bar{\omega} \lambda) - (\bar{\lambda} \omega) \\
& + M^* M + N^* N - V_\mu^* V^\mu.
\end{aligned} \tag{26.2.33}$$

The terms quadratic in C or ω look promising as the kinematic Lagrangians for massless fields of spins zero and 1/2; the final three terms are harmless; but the terms involving D or λ have the disastrous effect in path integrals of constraining C and ω to vanish. Fortunately, as we shall see in the following section, there are constrained superfields from which we can construct physically sensible actions. The introduction of these constrained superfields will also open up ways of constructing supersymmetric terms in the action that are not the D -components of functions of superfields.

If parity is conserved, then the space inversion properties of the component fields of a superfield will be related by supersymmetry. To work out this relation, we apply the parity operator P to the commutation/anticommutation relation (26.2.1) and use the transformation property (??) of supersymmetry generators, which gives

$$i\beta [Q, P^{-1}S(x, \theta)P] = \mathcal{Q}P^{-1}S(x, \theta)P. \tag{26.2.34}$$

The solution of Eq. (26.2.34) for a scalar superfield is of the form

$$P^{-1}S(x, \theta)P = \eta S(\Lambda_P x, -i\beta\theta), \tag{26.2.35}$$

where η is some phase (the intrinsic parity of the superfield) and $\Lambda_P x \equiv (-\mathbf{x}, x^0)$. (To check that Eq. (26.2.35) satisfies Eq. (26.2.34), note that Eq. (26.2.35) gives the left-hand side of Eq. (26.2.34) as

$$i\eta\beta \left(-\frac{\partial}{\partial(-i\beta\theta)} + \gamma^\mu(-i\beta\theta)\frac{\partial}{\partial(\Lambda_P x)^\mu} \right) S(\Lambda_P x, \theta) = \eta\mathcal{Q}S(\Lambda_P x, -i\beta\theta),$$

in agreement with what Eq. (26.2.35) gives for the right-hand side of Eq. (26.2.34).) Using the expansion (26.2.10) in Eq. (26.2.35) then gives the space inversion properties of the component fields:

$$\begin{aligned}
P^{-1}C(x)P &= \eta C(\Lambda_P x), \\
P^{-1}\omega(x)P &= -i\eta\beta\omega(\Lambda_P x), \\
P^{-1}M(x)P &= -\eta M(\Lambda_P x), \\
P^{-1}N(x)P &= \eta N(\Lambda_P x), \\
P^{-1}V^\mu(x)P &= -\eta(\Lambda_P)^\mu{}_\nu V^\nu(\Lambda_P x), \\
P^{-1}\lambda(x)P &= i\eta\beta\lambda(\Lambda_P x), \\
P^{-1}D(x)P &= \eta D(\Lambda_P x).
\end{aligned} \tag{26.2.36}$$

* * *

The general real superfield S involves four real spinless fields C , M , N , and D , plus one real four-vector field V_μ , for a total of eight independent bosonic field components. For comparison, there are two four-component Majorana spinor fields ω and λ , also for a total of eight independent field components. The equality of the numbers of independent bosonic and fermionic field components holds in general not only for the unconstrained general superfield studied in this section, but also for all superfields obtained from the general superfield by imposing supersymmetric constraints, such as the chiral and other constrained superfields discussed in the next section.

To see this in general, suppose that we have a representation of the supersymmetry algebra provided by N_B linearly independent real bosonic field operators $b_n(x)$ and N_F linearly independent fermionic field operators $f_k(x)$. We will assume that these fields satisfy only non-trivial field equations, so that no linear combination of the b_n or f_k with non-vanishing coefficients satisfies a homogeneous linear field equation. Consider a real supersymmetry generator $Q(u)$, defined as

$$Q(u) \equiv (\bar{u}Q) = (\bar{Q}u), \quad (26.2.37)$$

where u is some ordinary numerical Majorana spinor (*not* an anticommuting c-number). (For extended supersymmetry, in place of Q_α we could use any one of the $Q_{r\alpha}$, say $Q_{1\alpha}$.) In order for the b_n and f_k to furnish a representation of the supersymmetry algebra, we must have

$$[Q(u), b_n] = i \sum_k q_{nk}(\partial) f_k, \quad (26.2.38)$$

$$\{Q(u), f_k\} = \sum_n p_{kn}(\partial) b_n, \quad (26.2.39)$$

for some matrix differential operators $q(\partial)$ and $p(\partial)$. Taking the anticommutator of Eq. (26.2.38) and the commutator of Eq. (26.2.39) with $Q(u)$ gives

$$[Q^2(u), b_n] = i \sum_m (q(\partial)p(\partial))_{nm} b_m, \quad (26.2.40)$$

$$[Q^2(u), f_k] = i \sum_\ell (p(\partial)q(\partial))_{k\ell} f_\ell. \quad (26.2.41)$$

The anticommutation relation (??) or (??) gives the square of $Q(u)$ as $Q^2(u) = -iP_\mu(\bar{u}\gamma^\mu u)$. Hence the square matrices $p(\partial)q(\partial)$ and $q(\partial)p(\partial)$ must both be non-singular, because if there were any non-vanishing coefficients $c_n(\partial)$ or $d_k(\partial)$ for which $\sum_n c_n(\partial)(q(\partial)p(\partial))_{nm} = 0$ or $\sum_k d_k(\partial)(p(\partial)q(\partial))_{k\ell} = 0$ then b_n or f_k would satisfy the homogeneous linear field equations

$$(\bar{u}\gamma^\mu u)\partial_\mu \sum_n c_n(\partial) b_n = 0 \quad \text{or} \quad (\bar{u}\gamma^\mu u)\partial_\mu \sum_k d_k(\partial) f_k = 0,$$

in contradiction with our assumption that the fields do not satisfy such field equations. In order for qp to be non-singular we must have $N_F \geq N_B$, and in order for pq to be non-singular we must have $N_B \geq N_F$, so we can conclude that $N_B = N_F$. Also the square

matrices q and p must both be non-singular, so the complex conjugate of Eq. (26.2.38) tells us that $f^* = q^{*-1} q f$, so that the number of independent fermion fields is N_F rather than $2N_F$, and is therefore equal to the number N_B of independent boson fields, as was to be shown.

26.3 Chiral and Linear Superfields

In the previous section we found that the presence of D and λ components in a general superfield stood in the way of using such superfields in a physically satisfactory Lagrangian density. Suppose then that we consider a superfield with

$$\lambda = D = 0. \quad (26.3.1)$$

Are these conditions preserved by supersymmetry transformations? According to Eqs. (26.2.17) and (26.2.16), the condition $D = 0$ is invariant if $\lambda = 0$, but the condition $\lambda = 0$ is only invariant if we also impose the condition that $\partial_\mu V_\nu - \partial_\nu V_\mu = 0$, which requires that V_μ be a pure gauge:

$$V_\mu(x) = \partial_\mu Z(x). \quad (26.3.2)$$

Eq. (26.2.15) shows that, with $\lambda = 0$, this condition is preserved by a supersymmetry transformation. We have thus arrived at a reduced superfield, subject to the constraints (26.3.1) and (26.3.2), with component fields having the transformation properties

$$\delta C = i(\bar{\alpha} \gamma_5 \omega), \quad (26.3.3)$$

$$\delta \omega = (-i\gamma_5 \not{\partial} C - M + i\gamma_5 N + \not{\partial} Z) \alpha, \quad (26.3.4)$$

$$\delta M = -(\bar{\alpha} \not{\partial} \omega), \quad (26.3.5)$$

$$\delta N = i(\bar{\alpha} \gamma_5 \not{\partial} \omega), \quad (26.3.6)$$

$$\delta Z = (\bar{\alpha} \omega). \quad (26.3.7)$$

Comparing with Eq. (26.1.21), we see that this is the same as the supermultiplet constructed by direct methods in Section 26.1, with the identifications

$$C = A, \quad \omega = -i\gamma_5 \psi, \quad M = G, \quad N = -F, \quad Z = B. \quad (26.3.8)$$

A superfield satisfying the conditions (26.3.1) and (26.3.2) is said to be *chiral*.³

To distinguish a chiral superfield $X(x, \theta)$ from the general superfield $S(x, \theta)$ of the previous section, we will use A , B , F , G , and ψ for its components, instead of C , M , N ,

³Some authors use the term ‘chiral’ to describe a special case of such superfields, introduced below, which are here called left-chiral or right-chiral. Our use here of the term chiral may at first seem strange, because it has no counterpart for Dirac spinors. Any Dirac spinor is the sum of Dirac spinors that are left-chiral and right-chiral, in the sense that they are respectively proportional to $1 + \gamma_5$ and $1 - \gamma_5$, so no special term is needed for such sums of Dirac spinors. In contrast, it is only superfields satisfying Eqs. (26.3.1) and (26.3.2) that can be expressed as the sum of a left-chiral and a right-chiral superfield.

Z , and ω . By using Eqs. (26.3.1), (26.3.2), and (26.3.8) in Eq. (26.2.10), we find the form of a general chiral superfield to be

$$\begin{aligned} X(x, \theta) = & A(x) - (\bar{\theta}\psi(x)) + \frac{1}{2}(\bar{\theta}\theta)F(x) - \frac{i}{2}(\bar{\theta}\gamma_5\theta)G(x) \\ & + \frac{i}{2}(\bar{\theta}\gamma_5\gamma_\mu\theta)\partial^\mu B(x) + \frac{1}{2}(\bar{\theta}\gamma_5\theta)(\bar{\theta}\gamma_5\theta\psi(x)) \\ & - \frac{1}{8}(\bar{\theta}\gamma_5\theta)^2\Box A(x). \end{aligned} \quad (26.3.9)$$

(We could just as well have taken $C = -B$, $\omega = \psi$, $M = -F$, $N = -G$, and $Z = A$. We make the identifications (26.3.8) because, as we see here, for a scalar superfield they are consistent with the usual convention that A and F are scalars while B and G are pseudoscalars.)

The chiral superfield (26.3.9) may be further decomposed, as

$$X(x, \theta) = \frac{1}{\sqrt{2}} \left[\Phi(x, \theta) + \tilde{\Phi}(x, \theta) \right], \quad (26.3.10)$$

where

$$\begin{aligned} \Phi(x, \theta) = & \phi(x) - \sqrt{2}(\bar{\theta}\psi_L(x)) + \mathcal{F}(x) \left(\bar{\theta} \left(\frac{1+\gamma_5}{2} \right) \theta \right) + \frac{1}{2}(\bar{\theta}\gamma_5\gamma_\mu\theta)\partial^\mu\phi(x) \\ & - \frac{1}{\sqrt{2}}(\bar{\theta}\gamma_5\theta)(\bar{\theta}\theta\psi_L(x)) - \frac{1}{8}(\bar{\theta}\gamma_5\theta)^2\Box\phi(x), \end{aligned} \quad (26.3.11)$$

$$\begin{aligned} \tilde{\Phi}(x, \theta) = & \tilde{\phi}(x) - \sqrt{2}(\bar{\theta}\psi_R(x)) + \tilde{\mathcal{F}}(x) \left(\bar{\theta} \left(\frac{1-\gamma_5}{2} \right) \theta \right) - \frac{1}{2}(\bar{\theta}\gamma_5\gamma_\mu\theta)\partial^\mu\tilde{\phi}(x) \\ & + \frac{1}{\sqrt{2}}(\bar{\theta}\gamma_5\theta)(\bar{\theta}\theta\psi_R(x)) - \frac{1}{8}(\bar{\theta}\gamma_5\theta)^2\Box\tilde{\phi}(x), \end{aligned} \quad (26.3.12)$$

with component fields defined by

$$\phi \equiv \frac{A+iB}{\sqrt{2}}, \quad \psi_L \equiv \left(\frac{1+\gamma_5}{2} \right) \psi, \quad \mathcal{F} \equiv \frac{F-iG}{\sqrt{2}}, \quad (26.3.13)$$

$$\tilde{\phi} \equiv \frac{A-iB}{\sqrt{2}}, \quad \psi_R \equiv \left(\frac{1-\gamma_5}{2} \right) \psi, \quad \tilde{\mathcal{F}} \equiv \frac{F+iG}{\sqrt{2}}. \quad (26.3.14)$$

The component fields of either Φ or $\tilde{\Phi}$ furnish complete representations of the supersymmetry algebra:

$$\delta\psi_L = \sqrt{2}\partial_\mu\phi\gamma^\mu\alpha_R + \sqrt{2}\mathcal{F}\alpha_L, \quad (26.3.15)$$

$$\delta\mathcal{F} = \sqrt{2}(\bar{\alpha}_L\theta\psi_L), \quad (26.3.16)$$

$$\delta\phi = \sqrt{2}(\bar{\alpha}_R\psi_L), \quad (26.3.17)$$

$$\delta\psi_R = \sqrt{2}\partial_\mu\tilde{\phi}\gamma^\mu\alpha_L + \sqrt{2}\tilde{\mathcal{F}}\alpha_R, \quad (26.3.18)$$

$$\delta\tilde{\mathcal{F}} = \sqrt{2}(\bar{\alpha}_R\theta\psi_R), \quad (26.3.19)$$

$$\delta\tilde{\phi} = \sqrt{2}(\bar{\alpha}_L\psi_R), \quad (26.3.20)$$

where as usual

$$\alpha_L = \left(\frac{1 + \gamma_5}{2} \right) \alpha, \quad \alpha_R = \left(\frac{1 - \gamma_5}{2} \right) \alpha,$$

and likewise for θ . A superfield of the form (26.3.11) or (26.3.12) is known as *left-chiral* or *right-chiral*, respectively. In the special case where a chiral superfield $X(x, \theta)$ is real, its left-chiral and right-chiral parts Φ and $\tilde{\Phi}$ are complex conjugates, so that $\tilde{\phi} = \phi^*$, $\tilde{\mathcal{F}} = \mathcal{F}^*$, and ψ is a Majorana field. However, if we do not require $X(x, \theta)$ to be real then in general there is no relation between Φ and $\tilde{\Phi}$; it is even possible that one of the two vanishes.

The component fields of the superfield Φ include two complex bosonic components ϕ and $\mathcal{F}$, or four independent real bosonic components, and one Majorana fermion field ψ , which has four independent fermionic components. This is another example of the general result, derived at the end of the previous section, that any set of fields that furnish a representation of the supersymmetry algebra must have an equal number of independent bosonic and fermionic components.

We can use Eqs. (26.A.5), (26.A.17), and (26.A.18) to rewrite Eqs. (26.3.11) and (26.3.12) in a form that clarifies the way that these superfields depend on θ_L and θ_R :

$$\Phi(x, \theta) = \phi(x_+) - \sqrt{2}(\theta_L^T \epsilon \psi_L(x_+)) + \mathcal{F}(x_+)(\theta_L^T \epsilon \theta_L), \quad (26.3.21)$$

$$\tilde{\Phi}(x, \theta) = \tilde{\phi}(x_-) + \sqrt{2}(\theta_R^T \epsilon \psi_R(x_-)) - \tilde{\mathcal{F}}(x_-)(\theta_R^T \epsilon \theta_R), \quad (26.3.22)$$

where

$$x_{\pm}^{\mu} \equiv x^{\mu} \pm \frac{1}{2}(\bar{\theta} \gamma_5 \gamma^{\mu} \theta) = x^{\mu} \pm (\theta_R^T \epsilon \gamma^{\mu} \theta_L). \quad (26.3.23)$$

The expansions of $\phi(x_+)$ and $\tilde{\phi}(x_-)$ in powers of $x^{\mu} - x_{\pm}^{\mu}$ terminate with quadratic terms, the expansions of $\psi_{L,R}(x_{\pm})$ terminate with linear terms, while the expansions of $\mathcal{F}(x_+)$ and $\tilde{\mathcal{F}}(x_-)$ terminate at zeroth order, because all higher terms make contributions in Eqs. (26.3.21) and (26.3.22) that contain three or more factors of θ_L or θ_R and therefore vanish. For the same reason, it is easy to see that any superfield that depends only on θ_L and x_+^{μ} but not otherwise on θ_R must take the form (26.3.21), and any superfield that depends only on θ_R and x_-^{μ} but not otherwise on θ_L must take the form (26.3.22).

We have seen that for a superfield to be left-chiral or right-chiral is entirely a matter of what the superfield is allowed to depend on. It follows immediately that *any function of left-chiral superfields (or of right-chiral superfields), but not their complex conjugates or spacetime derivatives, is a left-(or right-)chiral superfield*. This can also be shown in a more formal way. Because $\Phi(x, \theta)$ depends on θ_R only through its dependence on x_+ , and $\tilde{\Phi}(x, \theta)$ depends on θ_L only through its dependence on x_- , they satisfy the conditions

$$\mathcal{D}_{R\alpha} \Phi = \mathcal{D}_{L\alpha} \tilde{\Phi} = 0, \quad (26.3.24)$$

where $\mathcal{D}_R$ and $\mathcal{D}_L$ are the right- and left-handed parts of the superderivative (26.2.26):

$$\mathcal{D}_{R\alpha} \equiv \left[\left(\frac{1 - \gamma_5}{2} \right) \mathcal{D} \right]_{\alpha} = - \sum_{\beta} \epsilon_{\alpha\beta} \frac{\partial}{\partial \theta_{R\beta}} - (\gamma^{\mu} \theta_L)_{\alpha} \frac{\partial}{\partial x^{\mu}}, \quad (26.3.25)$$

$$\mathcal{D}_{L\alpha} \equiv \left[\left(\frac{1 + \gamma_5}{2} \right) \mathcal{D} \right]_{\alpha} = + \sum_{\beta} \epsilon_{\alpha\beta} \frac{\partial}{\partial \theta_{L\beta}} - (\gamma^{\mu} \theta_R)_{\alpha} \frac{\partial}{\partial x^{\mu}}, \quad (26.3.26)$$

for which

$$\mathcal{D}_{R\alpha} x_{+}^{\mu} = \mathcal{D}_{L\alpha} x_{-}^{\mu} = 0.$$

Conversely, if a superfield Φ satisfies $\mathcal{D}_R \Phi = 0$ it is left-chiral, and if it satisfies $\mathcal{D}_L \Phi = 0$ it is right-chiral. Any function $f(\Phi)$ of superfields Φ_n that all satisfy $\mathcal{D}_R \Phi_n = 0$ or all satisfy $\mathcal{D}_L \Phi_n = 0$ will satisfy $\mathcal{D}_R f(\Phi) = 0$ or $\mathcal{D}_L f(\Phi) = 0$, and hence be left-chiral or right-chiral, respectively. But a function of left-chiral and right-chiral superfields is in general not chiral at all.

Using the representation (26.3.21) for left-chiral superfields makes it easy to work out their multiplication properties. For instance, if Φ_1 and Φ_2 are two left-chiral superfields, then their product $\Phi = \Phi_1 \Phi_2$ is a left-chiral superfield, with components

$$\phi = \phi_1 \phi_2, \quad (26.3.27)$$

$$\psi_L = \phi_1 \psi_{2L} + \phi_2 \psi_{1L}, \quad (26.3.28)$$

$$\mathcal{F} = \phi_1 \mathcal{F}_2 + \phi_2 \mathcal{F}_1 - (\psi_{1L}^T \epsilon \psi_{2L}). \quad (26.3.29)$$

The presence of chiral superfields in a theory opens up an additional possibility for constructing supersymmetric actions. Inspection of the transformation rule (26.3.16) shows that a supersymmetry transformation changes the $\mathcal{F}$ -term of a left-chiral superfield Φ by a derivative, so that the integral of the $\mathcal{F}$ -term of any left-chiral superfield is supersymmetric. Thus we can form a supersymmetric action as

$$I = \int d^4x [f]_{\mathcal{F}} + \int d^4x [f]_{\mathcal{F}}^* + \frac{1}{2} \int d^4x [K]_D, \quad (26.3.30)$$

where f and K are any left-chiral superfield and general real superfield, respectively, formed from the elementary superfields.

On what can f and K depend? The function f will be left-chiral if it depends only on left-chiral elementary superfields Φ_n , but not their right-chiral complex conjugates. On the other hand, the superderivative of a chiral superfield is not chiral, so we cannot freely include superderivatives of the Φ_n in f . It is true that, acting on a superfield S that is not left-chiral (such as one involving complex conjugates of left-chiral superfields), a pair of right superderivatives gives a left-chiral superfield, because there are only two independent right superderivatives and they anticommute:

$$\mathcal{D}_{R\alpha} (\mathcal{D}_{R\beta} \mathcal{D}_{R\gamma} S) = 0.$$

However, the $\mathcal{F}$ -term of any function f that is constructed in this way makes a contribution to the action that is the same as that of a D term of some other composite superfield. Since the $\mathcal{D}$ s anticommute, the most general left-chiral superfield formed by acting on a general superfield S with two $\mathcal{D}_R$ s may be expressed in terms of $(\mathcal{D}_R^T \epsilon \mathcal{D}_R) S$. If one of the left-chiral superfields in the superpotential is of this form, then since each $\mathcal{D}_R$ annihilates all the other

superfields in the superpotential, we can write the whole superpotential as $f = (\mathcal{D}_R^T \epsilon \mathcal{D}_R)h$ for some other superfield h . Now,

$$(\mathcal{D}_R^T \epsilon \mathcal{D}_R)(\theta_R^T \epsilon \theta_R) = -4,$$

so, apart from spacetime derivatives that do not contribute to the action, $(\mathcal{D}_R^T \epsilon \mathcal{D}_R)h$ is the coefficient of $-(\theta_R^T \epsilon \theta_R)/4$ in h . But, again apart from spacetime derivatives, $[f]_{\mathcal{F}}$ is the coefficient of $(\theta_L^T \epsilon \theta_L)$ in f , so $[(\mathcal{D}_R^T \epsilon \mathcal{D}_R)h]_{\mathcal{F}}$ equals the coefficient of

$$-\frac{1}{4}(\theta_L^T \epsilon \theta_L)(\theta_R^T \epsilon \theta_R) = -\frac{1}{4}(\bar{\theta} \gamma_5 \theta)^2$$

in h , and therefore

$$\int d^4x [(\mathcal{D}_R^T \epsilon \mathcal{D}_R)h]_{\mathcal{F}} = 2 \int d^4x [h]_D. \quad (26.3.31)$$

Thus we do not need to include terms in f that depend on left-chiral superfields of the form $\mathcal{D}_{R\beta} \mathcal{D}_{R\gamma} S$ —any such terms will be included in the list of all possible D -terms. When f is expressed as a function only of elementary left-chiral superfields and not their superderivatives or spacetime derivatives, it is known as the *superpotential*.

In contrast, the function K is in general a real scalar function of both left-chiral superfields Φ_n and their right-chiral complex conjugates Φ_n^* , as well as their superderivatives and spacetime derivatives, known as the *Kahler potential*. (Any right-chiral superfield is the complex conjugate of a left-chiral superfield, so there is no loss of generality in supposing K to depend only on left-chiral superfields and their complex conjugates.) However, not all K s obtained in this way will yield distinct actions. For one thing, chiral superfields have no D -terms, so two K s that differ by a chiral superfield make the same contribution to the action.

It is also possible to change the form of K without changing the action by a partial integration in superspace. The D -term of the superderivative $\mathcal{D}_\alpha S$ of an arbitrary superfield makes no contribution to the action because

$$\int d^4x [\mathcal{D}_\alpha S]_D = 0. \quad (26.3.32)$$

To see this, recall that

$$\mathcal{D}_\alpha S = \sum_\beta (\gamma_5 \epsilon)_{\alpha\beta} \frac{\partial S}{\partial \theta_\beta} - (\gamma^\mu \theta)_\alpha \frac{\partial S}{\partial x^\mu}.$$

Since S is a polynomial at most of fourth order in θ , the first term in $\mathcal{D}_\alpha S$ is a polynomial in θ at most of third order and therefore cannot have a non-zero D -term that is not a derivative, while the second term is also a spacetime derivative, so that its D -term is also a spacetime derivative, and therefore neither the first nor the second terms in $\mathcal{D}_\alpha S$ can contribute to the integral in Eq. (26.3.32). Also, the superderivative acts distributively, so it follows from Eq. (26.3.32) that we can integrate by parts in superspace: for any two bosonic superfields S_1 and S_2 ,

$$\int d^4x [S_1 \mathcal{D}_\alpha S_2]_D = - \int d^4x [S_2 \mathcal{D}_\alpha S_1]_D. \quad (26.3.33)$$

In Sections 26.4 and 26.8 we will consider in detail the case where f and K depend only on elementary superfields, but not their superderivatives or ordinary derivatives.

We saw in the previous section that in theories in which parity is conserved, the effect of the space inversion operator on a general scalar superfield is to subject its arguments to the transformations $x^\mu \rightarrow (\Lambda_P)^\mu{}_\nu x^\nu$ and $\theta \rightarrow -i\beta\theta$, and perhaps to multiply the superfield by a phase η . Under these transformations, the arguments $x^\mu_\pm$ in Eqs. (26.3.21) and (26.3.22) are changed by

$$x^\mu_\pm \longrightarrow (\Lambda_P x)^\mu \pm \frac{1}{2}(\bar{\theta}\beta\gamma_5\gamma^\mu\beta\theta) = (\Lambda_P x_\mp)^\mu, \quad (26.3.34)$$

while $\theta_L \rightarrow -i\beta\theta_R$ and $\theta_R \rightarrow -i\beta\theta_L$. Thus space inversion takes left-chiral superfields into right-chiral superfields, and vice-versa. The only right-handed scalar superfield whose component fields involve creation and annihilation operators for the same particles that are created and destroyed by a left-handed scalar superfield Φ is $\tilde{\Phi} \propto \Phi^*$, so $P^{-1}\Phi P$ must be proportional to Φ^* . By a suitable choice of phase of Φ , we can arrange that this transformation rule reads

$$P^{-1}\Phi(x, \theta)P = \Phi^*(\Lambda_P x, -i\beta\theta). \quad (26.3.35)$$

In terms of component fields, this transformation is

$$\begin{aligned} P^{-1}\phi(x)P &= \phi^*(\Lambda_P x), \\ P^{-1}\psi_L(x)P &= -i\epsilon\gamma_5\beta\psi_L^*(\Lambda_P x), \\ P^{-1}\mathcal{F}(x)P &= \mathcal{F}^*(\Lambda_P x). \end{aligned} \quad (26.3.36)$$

There is another type of possible symmetry, known as R -symmetry, that is important in some models of spontaneous supersymmetry breaking discussed in Section 26.5, and that will also be used in proving no-renormalization theorems in Section 27.6. As mentioned in Section 25.2, in theories of simple $N = 1$ supersymmetry an R -symmetry is invariance under a $U(1)$ transformation under which the left-handed components of the supersymmetry generator (called Q_a in Section 25.2) carry a non-vanishing quantum number, say -1 , in which case their adjoints, the right-handed components of the supersymmetry generator carry the opposite quantum number $+1$. Inspection of Eq. (26.2.2) shows that the θ superspace coordinate has a non-trivial transformation property under R -symmetry transformations: θ_L carries R -quantum number $+1$ and θ_R , which is proportional to θ_L^* , carries R quantum number -1 . In addition, the whole superfield may be given an R quantum number. If we give a left-chiral superfield Φ the R quantum number R_Φ , then its scalar component ϕ has the same R quantum number, while the left-handed spinor component ψ_L has $R_\psi = R_\Phi - 1$, and the auxiliary field $\mathcal{F}$ has $R_\mathcal{F} = R_\Phi - 2$. In particular, in order for the superpotential term $\int d^4x [f]_\mathcal{F}$ to conserve R , the superpotential itself must have $R_f = +2$, so if f depends on a single left-chiral superfield Φ , then it must be proportional to Φ^{2/R_Φ} . To put this another way, if $f(\Phi)$ is a pure mass term proportional to Φ^2 then we must choose $R_\Phi = +1$, while if $f(\Phi)$ is a pure interaction term proportional to Φ^3 , then we must choose $R_\Phi = 2/3$. On the other hand, inspection of Eq. (26.2.10) shows that the D -term of a superfield has the same R value as the superfield, so in order for the term

$\int d^4x [g]_D$ in the action to conserve R it is only necessary that K have $R = 0$, which will be the case if each term in K contains equal numbers of Φ and Φ^* factors, whatever R value we give to Φ . Of course, there is no general reason why R -symmetry should be respected by the action, or why it should not be spontaneously broken.

* * *

There are other ways of constraining superfields, to yield other types of supermultiplets of fields. Among the more common are the *linear* superfields. To learn the conditions defining this sort of superfield, we note that if S is a general superfield, then we can form a chiral superfield

$$S' \equiv \frac{1}{4}(\bar{\mathcal{D}}\mathcal{D})S. \quad (26.3.37)$$

This is a chiral superfield because it can be written as a sum of $\frac{1}{4}(\bar{\mathcal{D}}_L\mathcal{D}_L)S$, which is right-chiral, and $\frac{1}{4}(\bar{\mathcal{D}}_R\mathcal{D}_R)S$, which is left-chiral. Its components are given in terms of the components of S by

$$C' = N, \quad (26.3.38)$$

$$\omega' = \lambda + \not{\partial}\omega, \quad (26.3.39)$$

$$M' = -\partial_\mu V^\mu, \quad (26.3.40)$$

$$N' = D + \square C, \quad (26.3.41)$$

$$V'_\mu = -\partial_\mu M, \quad (26.3.42)$$

$$\lambda' = D' = 0. \quad (26.3.43)$$

A multiplet S is said to be linear if the superfield S' defined in this way vanishes

$$(\bar{\mathcal{D}}\mathcal{D})S = 0, \quad (26.3.44)$$

or in terms of its components,

$$N = M = \partial_\mu V^\mu = 0, \quad \lambda = -\not{\partial}\omega, \quad D = -\square C. \quad (26.3.45)$$

This leaves four independent bosonic fields, C and the three components of V_μ subject to the condition $\partial_\mu V^\mu = 0$, and four independent fermionic fields, the components of the Majorana four-spinor ω . We will see in Section 26.6 that the current superfields, whose V_μ -terms are the conserved currents associated with symmetry transformations, are linear superfields.

26.4 Renormalizable Theories of Chiral Superfields

We will now work out the details of a general renormalizable theory of scalar chiral superfields. This will provide some insight into the implications of supersymmetry, and the theory we obtain will be part of the supersymmetric standard model to be discussed in Chapter 28.

As discussed in Section 12.2, the Lagrangian density in a renormalizable theory can contain only operators with dimensionality (counting powers of energy or momentum, with $\hbar = c = 1$) four or less. Eq. (26.2.6) shows that $\mathcal{Q}_\alpha$ and hence $\partial/\partial\theta_\alpha$ has dimensionality $1/2$, so $\mathcal{D}_\alpha$ has dimensionality $+1/2$, and θ_α has dimensionality $-1/2$. The $\mathcal{F}$ - and D -terms of a superfield S are the coefficients of two or four factors of θ , respectively, so if the superfield has dimensionality $d(S)$ then its $\mathcal{F}$ - and D -terms have dimensionalities $d(\mathcal{F}^S) = d(S) + 1$ and $d(D^S) = d(S) + 2$. Thus in a renormalizable theory the functions f and K in Eq. (26.3.30) consist of operators with dimensionality at most three and two, respectively.

The dimensionality of an elementary scalar superfield Φ_n is that of an elementary scalar field, or $+1$, so in order for each term in the function f to have dimensionality three or less, it can contain at most three factors of Φ_n and/or derivatives $\partial/\partial x^\mu$ and/or pairs of spinor superderivatives $\mathcal{D}_\alpha$. As discussed in the previous section, any left-chiral term in f that involves superderivatives could be replaced with a term in K , so superderivatives can be omitted in f . Eq. (26.2.30) shows that spacetime derivatives can be expressed in terms of superderivatives, so these too can be omitted. (In any case, Lorentz invariance would rule out terms with one spacetime derivative, and terms with two derivatives in a renormalizable theory could involve only one Φ_n factor, on which these derivatives would have to act, so such terms would not contribute to the action.) We conclude that $f(\Phi)$ is at most a cubic polynomial in the Φ_n , without spacetime derivatives or superderivatives.

The same dimensional analysis shows that in a renormalizable theory K is at most a quadratic function of Φ_n and Φ_n^* , without derivatives. But any term in $K(\Phi, \Phi^*)$ that involves only Φ_n or only Φ_n^* would be a chiral superfield, and chiral superfields by definition have no D -terms, so $[K(\Phi, \Phi^*)]_D$ receives contributions only from terms in $K(\Phi, \Phi^*)$ that involve both Φ_n and Φ_n^* . Thus $K(\Phi, \Phi^*)$ must be of the form

$$K(\Phi, \Phi^*) = \sum_{mn} g_{nm} \Phi_n^* \Phi_m, \quad (26.4.1)$$

with constant coefficients g_{nm} forming a Hermitian matrix.

We must now calculate the $\mathcal{F}$ - and D -components of $f(\Phi)$ and $K(\Phi, \Phi^*)$, respectively. To calculate the D -component of $K(\Phi, \Phi^*)$, we note that the term in $\Phi_n^* \Phi_m$ of fourth order in θ is

$$\begin{aligned} [\Phi_n^* \Phi_m]_{\theta^4} = & -\frac{1}{8} (\bar{\theta} \gamma_5 \theta)^2 [\phi_n^* \square \phi_m + (\square \phi_m^*) \phi_n] \\ & + (\bar{\theta} \gamma_5 \theta) [(\bar{\psi}_n \theta) (\bar{\theta} \gamma^\mu \partial_\mu \psi_m) ((\partial_\mu \bar{\psi}_n) \gamma^\mu \theta) (\bar{\theta} \psi_m)] \\ & + \frac{1}{4} \mathcal{F}_n^* \mathcal{F}_m (\bar{\theta} (1 - \gamma_5) \theta) (\bar{\theta} (1 + \gamma_5) \theta) \\ & - \frac{1}{4} \partial^\mu \phi_n^* \partial^\nu \phi_m (\bar{\theta} \gamma_5 \gamma_\mu \theta) (\bar{\theta} \gamma_5 \gamma_\nu \theta). \end{aligned}$$

Using (26.A.18) and (26.A.19) allows us to convert the θ dependence of this expression to

an over-all factor $(\bar{\theta}\gamma_5\theta)^2$:

$$[\Phi_n^*\Phi_m]_{\theta^4} = -\frac{1}{4}(\bar{\theta}\gamma_5\theta)^2 \left[\frac{1}{2}\phi_n^*\square\phi_m + \frac{1}{2}(\square\phi_m^*)\phi_n - (\bar{\psi}_n\gamma^\mu\partial_\mu\psi_m) \right. \\ \left. + ((\partial_\mu\bar{\psi}_n)\gamma^\mu\psi_m) + 2\mathcal{F}_n^*\mathcal{F}_m - \partial^\mu\phi_n^*\partial_\mu\phi_m \right].$$

The D -term of a superfield is the coefficient of $-\frac{1}{4}(\bar{\theta}\gamma_5\theta)^2$ minus $\frac{1}{2}\square$ acting on the θ -independent term, which for $\Phi_n^*\Phi_m$ is $\phi_n^*\phi_m$, so

$$\frac{1}{2}[K(\Phi, \Phi^*)]_D = \sum_{nm} g_{nm} \left[-\partial_\mu\phi_n^*\partial^\mu\phi_m + \mathcal{F}_n^*\mathcal{F}_m \right. \\ \left. - \frac{1}{2}(\bar{\psi}_{nL}\gamma^\mu\partial_\mu\psi_{mL}) + \frac{1}{2}((\partial_\mu\bar{\psi}_{nL})\gamma^\mu\psi_{mL}) \right]. \quad (26.4.2)$$

If we write Φ_n as a linear combination $\sum_m N_{nm}\Phi'_m$ of new superfields Φ'_m , then $K(\Phi, \Phi^*)$ is given in terms of the new superfields by a formula which is the same as Eq. (26.4.1), except that g_{nm} is replaced with $g'_{nm} = (N^\dagger g N)_{nm}$. In order for the kinematic terms for the scalar and spinor fields to have a sign consistent with the quantum commutation and anticommutation relations, it is necessary that the Hermitian matrix g_{nm} be positive-definite and, as shown in Section 12.5, this means that we can choose N so that $g'_{nm} = \delta_{nm}$. Dropping primes, the term (26.4.2) is now

$$\frac{1}{2}[K(\Phi, \Phi^*)]_D = \sum_n \left[-\partial_\mu\phi_n^*\partial^\mu\phi_n + \mathcal{F}_n^*\mathcal{F}_n \right. \\ \left. - \frac{1}{2}(\bar{\psi}_{nL}\gamma^\mu\partial_\mu\psi_{nL}) + \frac{1}{2}((\partial_\mu\bar{\psi}_{nL})\gamma^\mu\psi_{nL}) \right]. \quad (26.4.3)$$

We can still use a unitary transformation to redefine the superfields without changing the form of Eq. (26.4.3), a freedom we will need to exercise shortly.

The terms in Eq. (26.4.3) involving ϕ_n and ψ_{nL} are the correct kinematic Lagrangians for conventionally normalized complex scalar and Majorana spinor fields. We will rewrite the fermion terms in a more familiar form after we have had a chance to consider mass terms.

To calculate the $\mathcal{F}$ -term of $f(\Phi)$, it is most convenient to use the representation (26.3.21) of the superfields, and pick out the term of second order in θ_L :

$$[f(\Phi(x, \theta))]_{\theta_L^2} = \sum_{nm} (\theta_L^T \epsilon \psi_{nL}(x)) (\theta_L^T \epsilon \psi_{mL}(x)) \frac{\partial^2 f(\phi(x))}{\partial\phi_n(x)\partial\phi_m(x)} \\ + \sum_n \mathcal{F}_n(x) \frac{\partial f(\phi(x))}{\partial\phi_n(x)} (\theta_L^T \epsilon \theta_L).$$

(We have replaced x_+ here with x because the term $(\theta_R^T \epsilon \gamma^\mu \theta_L)$ in Eq. (26.3.21) vanishes when multiplied by an expression with two θ_L factors.) The θ dependence of the first term

on the right-hand side can be put into a standard form by using Eq. (26.A.11) to write⁴

$$\begin{aligned} (\theta_L^T \epsilon \psi_{nL})(\theta_L^T \epsilon \psi_{mL}) &= \left(\psi_{nL}^T \epsilon \left(\frac{1 + \gamma_5}{2} \right) \theta \right) \left(\theta^T \epsilon \left(\frac{1 + \gamma_5}{2} \right) \psi_{mL} \right) \\ &= -\frac{1}{2} (\bar{\psi}_{nL} \psi_{mL}) (\theta_L^T \epsilon \theta_L). \end{aligned}$$

The $\mathcal{F}$ -term of any left-chiral superfield is the coefficient of $(\theta_L^T \epsilon \theta_L)$, so here

$$[f(\Phi)]_{\mathcal{F}} = -\frac{1}{2} \sum_{nm} \frac{\partial^2 f(\phi_n)}{\partial \phi_n \partial \phi_m} (\bar{\psi}_{nL} \psi_{mL}) + \sum_n \mathcal{F}_n \frac{\partial f(\phi)}{\partial \phi_n}. \quad (26.4.4)$$

The complete Lagrangian density is the sum of the terms (26.4.3), (26.4.4), and the complex conjugate of (26.4.4):

$$\begin{aligned} \mathcal{L} = \sum_n & \left[-\partial_\mu \phi_n^* \partial^\mu \phi_n + \mathcal{F}_n^* \mathcal{F}_n - \frac{1}{2} (\bar{\psi}_{nL} \gamma^\mu \partial_\mu \psi_{nL}) + \frac{1}{2} ((\partial_\mu \bar{\psi}_{nL}) \gamma^\mu \psi_{nL}) \right] \\ & - \frac{1}{2} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\bar{\psi}_{nL} \psi_{mL}) - \frac{1}{2} \sum_{nm} \left(\frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \right)^* (\bar{\psi}_{nL} \psi_{mL})^* \\ & + \sum_n \mathcal{F}_n \frac{\partial f(\phi)}{\partial \phi_n} + \sum_n \mathcal{F}_n^* \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^*. \end{aligned} \quad (26.4.5)$$

The auxiliary fields $\mathcal{F}_n$ enter quadratically in the action, with constant coefficients for the second-order terms, so they can be eliminated by setting $\mathcal{F}_n$ equal to the value at which the Lagrangian density (26.4.5) is stationary with respect to $\mathcal{F}_n$ and $\mathcal{F}_n^*$:

$$\mathcal{F}_n = - \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^*. \quad (26.4.6)$$

Inserting this in Eq. (26.4.5) gives

$$\begin{aligned} \mathcal{L} = \sum_n & \left[-\partial_\mu \phi_n^* \partial^\mu \phi_n - \frac{1}{2} (\bar{\psi}_{nL} \gamma^\mu \partial_\mu \psi_{nL}) + \frac{1}{2} ((\partial_\mu \bar{\psi}_{nL}) \gamma^\mu \psi_{nL}) \right] \\ & - \frac{1}{2} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\bar{\psi}_{nL} \psi_{mL}) - \frac{1}{2} \sum_{nm} \left(\frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \right)^* (\bar{\psi}_{nL} \psi_{mL})^* \\ & - \sum_n \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^* \frac{\partial f(\phi)}{\partial \phi_n}. \end{aligned} \quad (26.4.7)$$

Thus the scalar field potential is $V(\phi) = \sum_n |\partial f(\phi)/\partial \phi_n|^2$.

With the auxiliary fields eliminated in this way, the action is no longer invariant under the supersymmetry transformations (26.3.15), (26.3.17) of the remaining fields ψ_{nL} and ϕ_n :

$$\delta \psi_{nL} = \sqrt{2} \partial_\mu \phi_n \gamma^\mu \alpha_R - \sqrt{2} \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^* \alpha_L, \quad \delta \phi_n = \sqrt{2} (\bar{\alpha}_R \psi_{nL}).$$

⁴Note that $\bar{\psi}_{nL}$ is the left-handed component of $\bar{\psi}_n$, not $\overline{\psi_{nL}}$.

This is because the expression (26.4.6) does not obey the transformation rule $\delta\mathcal{F}_n = \sqrt{2}(\bar{\alpha}_L\not{\partial}\psi_{nL})$ for $\mathcal{F}_n$ given by Eq. (26.3.16), but instead

$$\begin{aligned}\delta\left(-\frac{\partial f(\phi)}{\partial\phi_n}\right)^* &= -\sum_m\left(\frac{\partial^2 f(\phi)}{\partial\phi_n\partial\phi_m}\right)^*\delta\phi_m^* \\ &= -\sqrt{2}\sum_m\left(\frac{\partial^2 f(\phi)}{\partial\phi_n\partial\phi_m}\right)^*(\bar{\alpha}_L\psi_{mR}).\end{aligned}$$

For the same reason, after eliminating the auxiliary fields the commutators of the supersymmetry transformations of ϕ_n and ψ_{nL} are no longer given by the supersymmetry anticommutation relations, and in fact do not form a closed Lie superalgebra. But this is not inconsistent with the existence of quantum mechanical operators Q_α that satisfy the anticommutation relations of supersymmetry. These operators generate supersymmetry transformations, in the sense that the commutator of $-i(\bar{\alpha}Q)$ with any Heisenberg-picture quantum field ϕ_n or ψ_{nL} equals the change in that field under a supersymmetry transformation with infinitesimal parameter α . With $\mathcal{F}_n$ given by Eq. (26.4.6) the commutator of $-i(\bar{\alpha}Q)$ with $\mathcal{F}_n$ is given by $\delta\mathcal{F}_n = \sqrt{2}(\bar{\alpha}_L\not{\partial}\psi_{nL})$, because in the Heisenberg picture the quantum field ψ_{nL} satisfies the field equation derived from the Lagrangian (26.4.7):

$$\not{\partial}\psi_{nL} = -\sum_m\left(\frac{\partial^2 f(\phi)}{\partial\phi_n\partial\phi_m}\right)^*\psi_{mR}.$$

Likewise, the supersymmetry transformations of the quantum fields ϕ_n and ψ_{nL} do form a closed Lie superalgebra when the field equations are taken into account. Such algebras are often called on-shell.

The zeroth-order expectation values ϕ_{n0} of the scalar fields ϕ_n must be at the maximum of the last term in Eq. (26.4.7). Since this term is always negative or zero, the maximum will be at spacetime-independent field values ϕ_{n0} at which this term vanishes, so that

$$\left.\frac{\partial f(\phi)}{\partial\phi_n}\right|_{\phi=\phi_0} = 0, \tag{26.4.8}$$

provided of course that a solution of this equation exists. Eq. (26.4.8) not only maximizes the last term in Eq. (26.4.7)—it is also the condition for supersymmetry to be unbroken. Invariance of the vacuum under supersymmetry transformations requires that the vacuum expectation value of the change in any field under a supersymmetry transformation should vanish. The change in a bosonic field is a fermionic field, which of course has zero expectation value anyway, but Eq. (26.3.15) shows that the vacuum expectation value of $\delta\psi_{nL}$ is proportional to the vacuum expectation value of the auxiliary field $\mathcal{F}_n$, which therefore must vanish if supersymmetry is unbroken. According to Eq. (26.4.6), in zeroth-order perturbation theory this condition requires that Eq. (26.4.8) must be satisfied. We will see in Section 27.6 that if Eq. (26.4.8) is satisfied then supersymmetry is unbroken to all orders of perturbation theory.

For a single left-chiral scalar superfield Φ , the fundamental theorem of algebra tells us that the polynomial $\partial f(\phi)/\partial\phi$ always has at least one zero somewhere in the complex

plane. This is not necessarily true when there is more than one superfield. If we assume that there is a solution ϕ_{n0} of Eq. (26.4.8), we can evaluate the physical degrees of freedom of the theory by setting

$$\phi_n = \phi_{n0} + \varphi_n, \quad (26.4.9)$$

and expanding in powers of φ_n . The masses of the particles of this theory can be calculated by inspecting the terms of second order in φ and ψ :

$$\begin{aligned} \mathcal{L}_0 = \sum_n \left[-\partial_\mu \varphi_n^* \partial^\mu \varphi_n - \frac{1}{2} (\bar{\psi}_{nL} \gamma^\mu \partial_\mu \psi_{nL}) + \frac{1}{2} ((\partial_\mu \bar{\psi}_{nL}) \gamma^\mu \psi_{nL}) \right] \\ - \frac{1}{2} \sum_{nm} \mathcal{M}_{nm} (\bar{\psi}_{nL} \psi_{mL}) - \frac{1}{2} \sum_{nm} \mathcal{M}_{nm}^* (\bar{\psi}_{nL} \psi_{mL})^* - \sum_{nm} (\mathcal{M}^\dagger \mathcal{M})_{mn} \varphi_m^* \varphi_n, \end{aligned} \quad (26.4.10)$$

where $\mathcal{M}$ is the symmetric complex matrix

$$\mathcal{M}_{mn} \equiv \left(\frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \right)_{\phi=\phi_0}. \quad (26.4.11)$$

Now if we redefine the fields by a unitary transformation

$$\varphi_n = \sum_m \mathcal{U}_{nm} \varphi'_m, \quad \psi_{nL} = \sum_m \mathcal{U}_{nm} \psi'_{mL}, \quad (26.4.12)$$

then the free-field Lagrangian (26.4.10) will take the same form, but with $\mathcal{M}$ replaced with $\mathcal{M}'$, where

$$\mathcal{M}' = \mathcal{U}^T \mathcal{M} \mathcal{U}. \quad (26.4.13)$$

According to a theorem of matrix algebra, for any complex symmetric matrix $\mathcal{M}$ it is always possible to find a unitary matrix $\mathcal{U}$ such that the matrix $\mathcal{M}'$ defined by Eq. (26.4.13) is diagonal, with real positive elements m_n on the diagonal. (For future use, we note that $\mathcal{M}'^\dagger \mathcal{M}' = \mathcal{U}^\dagger \mathcal{M}^\dagger \mathcal{M} \mathcal{U}$, so the quantities m_n^2 are just the eigenvalues of the positive Hermitian matrix $\mathcal{M}^\dagger \mathcal{M}$.) Redefining the fields in this way and dropping the primes, the quadratic part of the Lagrangian is now

$$\begin{aligned} \mathcal{L}_0 = \sum_n \left[-\partial_\mu \varphi_n^* \partial^\mu \varphi_n - \frac{1}{2} (\bar{\psi}_{nL} \gamma^\mu \partial_\mu \psi_{nL}) + \frac{1}{2} ((\partial_\mu \bar{\psi}_{nL}) \gamma^\mu \psi_{nL}) \right] \\ - \frac{1}{2} \sum_n m_n (\bar{\psi}_{nL} \psi_{nL}) - \frac{1}{2} \sum_n m_n (\bar{\psi}_{nL} \psi_{nL})^* - \sum_n m_n^2 \varphi_n^* \varphi_n. \end{aligned} \quad (26.4.14)$$

To put the fermion mass terms in a more familiar form, we introduce fields $\psi_n(x)$ defined as the Majorana fields whose left-handed components are $\psi_{nL}(x)$. Then, using the symmetry properties (26.A.7) of Majorana bilinears:

$$\begin{aligned} & -\frac{1}{2} (\bar{\psi}_{nL} \gamma^\mu \partial_\mu \psi_{nL}) + \frac{1}{2} ((\partial_\mu \bar{\psi}_{nL}) \gamma^\mu \psi_{nL}) \\ &= -\frac{1}{2} \left(\bar{\psi}_n \gamma^\mu \left(\frac{1+\gamma_5}{2} \right) \partial_\mu \psi_n \right) + \frac{1}{2} \left((\partial_\mu \bar{\psi}_n) \gamma^\mu \left(\frac{1+\gamma_5}{2} \right) \psi_n \right) \\ &= -\frac{1}{2} \left(\bar{\psi}_n \gamma^\mu \left(\frac{1+\gamma_5}{2} \right) \partial_\mu \psi_n \right) - \frac{1}{2} \left(\bar{\psi}_n \gamma^\mu \left(\frac{1-\gamma_5}{2} \right) \partial_\mu \psi_n \right) \\ &= -\frac{1}{2} (\bar{\psi}_n \gamma^\mu \partial_\mu \psi_n), \end{aligned}$$

while the reality properties (26.A.21) give

$$(\bar{\psi}_{nL}\psi_{nL}) + (\bar{\psi}_{nL}\psi_{nL})^* = 2\text{Re}\left(\bar{\psi}_n\left(\frac{1+\gamma_5}{2}\right)\psi_n\right) = (\bar{\psi}_n\psi_n).$$

The complete quadratic Lagrangian is then

$$\mathcal{L}_0 = \sum_n \left[-\partial_\mu \varphi_n^* \partial^\mu \varphi_n - \sum_n m_n^2 \varphi_n^* \varphi_n - \frac{1}{2}(\bar{\psi}_n \gamma^\mu \partial_\mu \psi_n) - \frac{m_n}{2}(\bar{\psi}_n \psi_n) \right]. \quad (26.4.15)$$

The factor 1/2 in the fermion terms is correct because these are Majorana fermion fields, while there is no factor 1/2 in the scalar terms because these are complex scalars. We see that the spinless and spin 1/2 particles have equal masses m_n , as required by the unbroken supersymmetry of the theory.

The interaction part $\mathcal{L}'$ of the Lagrangian density is given by the terms in Eq. (26.4.7) of higher than second order in φ_n and ψ_n . Since the superpotential $f(\phi_0 + \varphi)$ is being assumed to be a cubic polynomial and stationary at $\varphi_n = 0$, with φ defined so that the second-order terms are $\frac{1}{2} \sum_n m_n \varphi_n^2$, we may write the superpotential (apart from an inconsequential constant term) as

$$f(\phi_0 + \varphi) = \frac{1}{2} \sum_n m_n \varphi_n^2 + \frac{1}{6} \sum_{nm\ell} f_{nm\ell} \varphi_n \varphi_m \varphi_\ell. \quad (26.4.16)$$

Using this in Eq. (26.4.7) gives the interaction as

$$\begin{aligned} \mathcal{L}' = & -\frac{1}{2} \sum_{nm\ell} f_{nm\ell} \varphi_n \left(\bar{\psi}_m \left(\frac{1+\gamma_5}{2} \right) \psi_\ell \right) \\ & -\frac{1}{2} \sum_{nm\ell} f_{nm\ell}^* \varphi_n^* \left(\bar{\psi}_m \left(\frac{1-\gamma_5}{2} \right) \psi_\ell \right) \\ & -\frac{1}{2} \sum_{nm\ell} m_n f_{nm\ell} \varphi_n^* \varphi_m \varphi_\ell - \frac{1}{2} \sum_{nm\ell} m_n f_{nm\ell}^* \varphi_n \varphi_m^* \varphi_\ell^* \\ & -\frac{1}{4} \sum_{nm\ell m'\ell'} f_{nm\ell} f_{nm'\ell'}^* \varphi_m \varphi_\ell \varphi_{m'}^* \varphi_{\ell'}^*. \end{aligned} \quad (26.4.17)$$

We see that a knowledge of the masses m_n and the ‘Yukawa’ couplings $f_{nm\ell}$ of the scalars and fermions is enough to determine all the cubic and quartic self-couplings of the spinless fields.

As an illustration, consider the case of a single left-chiral superfield. For comparison with earlier results, let us write the single coefficient f in Eq. (26.4.16) as

$$f \equiv 2\sqrt{2} e^{i\alpha} \lambda, \quad (26.4.18)$$

where λ is real and α is some real phase. We will also introduce a pair of real spinless fields $A(x)$ and $B(x)$ by writing the single complex scalar here as

$$\varphi \equiv e^{-i\alpha} \left(\frac{A + iB}{\sqrt{2}} \right). \quad (26.4.19)$$

The total Lagrangian density is then given by the sum of Eqs. (26.4.15) and (26.4.17) as

$$\begin{aligned}\mathcal{L} = & -\frac{1}{2}\partial_\mu A \partial^\mu A - \frac{1}{2}\partial_\mu B \partial^\mu B - \frac{1}{2}m^2(A^2 + B^2) \\ & - \frac{1}{2}(\bar{\psi}\gamma^\mu\partial_\mu\psi) - \frac{1}{2}m(\bar{\psi}\psi) - \lambda A(\bar{\psi}\psi) - i\lambda B(\bar{\psi}\gamma_5\psi) \\ & - m\lambda A(A^2 + B^2) - \frac{1}{2}\lambda^2(A^2 + B^2)^2.\end{aligned}\tag{26.4.20}$$

which is the same as the Lagrangian density (??) originally found by Wess and Zumino [2]. It is noteworthy that in this simple case the Lagrangian turns out to be invariant under a space inversion transformation

$$A(x) \longrightarrow A(\Lambda_P x), \quad B(x) \longrightarrow -B(\Lambda_P x), \quad \psi(x) \longrightarrow i\beta\psi(\Lambda_P x), \tag{26.4.21}$$

even though we did not assume parity conservation in deriving it. The appearance of parity conservation as an ‘accidental’ symmetry is a familiar feature of various renormalizable gauge theories (see Sections 12.5 and 18.7) but not of theories involving spinless fields, so this is a special consequence of supersymmetry in the renormalizable theory of a single scalar superfield.

26.5 Spontaneous Supersymmetry Breaking in the Tree Approximation

We saw in the previous section that supersymmetry is unbroken (at least in the tree approximation) in renormalizable theories of chiral superfields if Eq. (26.4.8) has a solution, that is, if there is a value ϕ_0 of the fields at which the superpotential $f(\phi)$ is stationary:

$$\left. \frac{\partial f(\phi)}{\partial \phi_n} \right|_{\phi=\phi_0} = 0. \tag{26.5.1}$$

There are as many independent variables here as there are equations to satisfy, so we generally expect there to be solutions of Eq. (26.5.1). In order for supersymmetry to be spontaneously broken in these theories, it is necessary to impose restrictions on the form of the superpotential.

To see how a choice of superpotential may allow supersymmetry to be spontaneously broken, we will consider a generalization of a class of models due to O’Raifeartaigh [3]. Suppose that the superpotential is a linear combination of a set Y_i of left-chiral superfields, with coefficients given by functions $f_i(X)$ of a second set of left-chiral superfields X_n :

$$f(X, Y) = \sum_i Y_i f_i(X). \tag{26.5.2}$$

The conditions for supersymmetry to be unbroken by the values x_n and y_i of the scalar components of these superfields are that

$$0 = \frac{\partial f(x, y)}{\partial y_i} = f_i(x), \tag{26.5.3}$$

and

$$0 = \frac{\partial f(x, y)}{\partial x_n} = \sum_i y_i \frac{\partial f_i(x)}{\partial x_n}. \tag{26.5.4}$$

Eq. (26.5.4) can always be solved by taking $y_i = 0$, with no effect on the problem of solving Eq. (26.5.3). On the other hand, if the number of X_n superfields is smaller than the number of Y_i superfields, then Eq. (26.5.3) imposes more conditions on the x_n than there are variables, so without fine-tuning a solution is impossible, and supersymmetry is broken.

It may appear that the initial assumption (26.5.2) itself represents a radical form of fine-tuning, but in fact this form can be imposed on the superpotential by assuming a suitable R -symmetry. As discussed in Section 26.3, in theories with $N = 1$ supersymmetry an R -symmetry is a $U(1)$ symmetry for which the θ superspace coordinate has a non-trivial transformation property. If we assume an R -symmetry for which θ_L carries quantum number $+1$, then the $\mathcal{F}$ -term of any superpotential has a quantum number equal to that of the superpotential itself minus 2, so R invariance requires the superpotential itself to have $R = 2$. We can therefore impose the structure (26.5.2) by requiring R invariance, with the Y_i and X_n superfields given R quantum numbers $+2$ and 0 , respectively.

The scalar fields in this sort of model have a potential

$$V(x, y) = \sum_i |f_i(x)|^2 + \sum_n \left| \sum_i y_i \frac{\partial f_i(x)}{\partial x_n} \right|^2. \quad (26.5.5)$$

The potential is always minimized by choosing the x_n to minimize the first term; whatever values this gives the x_n , the second term can always be minimized by taking $y_i = 0$. Whether or not supersymmetry can be spontaneously broken, these models have the peculiar feature that there are always directions in the space of the fields in which the minimum of the potential is flat. Whatever values x_{n0} of the x_n minimize the first term in Eq. (26.5.5), the second term vanishes not only for $y_i = 0$, but for any vector y_i in a direction orthogonal to all the vectors $(v^n)_i = (\partial f_i(x)/\partial x_n)_{x=x_0}$. If there are N_X superfields X_n and N_Y superfields Y_i with $N_Y > N_X$, then the v^n cannot span the space of the y s, and there will be at least $N_Y - N_X$ of these flat directions. For any non-vanishing values $y_i = y_{0i}$ along any one of these flat directions, the R -symmetry of the Lagrangian density is spontaneously broken, and the Goldstone boson field ϕ associated with this global symmetry breakdown corresponds to a term $\propto y_{0i}$ in the y_i .

The simplest example of this class of models is provided by the case where there is just one X superfield and two Y superfields. Renormalizability requires the coefficient functions $f_i(X)$ to be quadratic functions of X , and by taking suitable linear combinations of the Y_i and shifting and rescaling X we can choose these functions so that

$$f_1(X) = X - a, \quad f_2(X) = X^2, \quad (26.5.6)$$

with an arbitrary constant a . There clearly is no simultaneous solution of the two equations (26.5.1) unless the superpotential is fine-tuned so that $a = 0$. The potential (26.5.5) here is

$$V(x, y) = |x|^4 + |x - a|^2 + |y_1 + 2xy_2|^2. \quad (26.5.7)$$

The sum of the first two terms has a unique global minimum x_0 . The flat direction here is the one for which $y_1 + 2x_0y_2 = 0$. For $a = 0$ we have $x_0 = 0$, and the minima of the potential are along the line with $y_1 = 0$ and y_2 arbitrary.

Whatever the reason for a spontaneous breakdown of supersymmetry, this phenomenon always entails the existence of a massless spin 1/2 particle, the *goldstino*, analogous to the Goldstone bosons associated with the spontaneous breakdown of ordinary global symmetries. (The one exception, discussed in Section 31.3, is that in supergravity theories, where supersymmetry is a local symmetry, the goldstino appears as the helicity $\pm 1/2$ states of a massive particle of spin 3/2, the gravitino.) In renormalizable theories of chiral superfields, the tree-approximation vacuum expectation values ϕ_{n0} of the scalar fields must be at a minimum of the potential $\sum_n |\partial f(\phi)/\partial \phi_n|^2$ in Eq. (26.4.7), so

$$\sum_m \mathcal{M}_{nm} \left(\frac{\partial f(\phi)}{\partial \phi_m} \Big|_{\phi=\phi_0} \right)^* = 0, \quad (26.5.8)$$

where

$$\mathcal{M}_{nm} \equiv \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \Big|_{\phi=\phi_0}. \quad (26.5.9)$$

If Eq. (26.5.1) is not satisfied, then Eq. (26.5.8) tells us that the matrix $\mathcal{M}_{nm}$ has at least one eigenvector with eigenvalue zero, so according to Eq. (26.4.10) there must be at least one linear combination of the spin 1/2 particles described by ψ_n with zero mass. For instance, for the model defined by Eqs. (26.5.2) and (26.5.6), the matrix $\mathcal{M}$ has the non-vanishing components

$$\mathcal{M}_{xy_1} = \mathcal{M}_{y_1x} = 1, \quad \mathcal{M}_{xy_2} = \mathcal{M}_{y_2x} = 2x_0, \quad (26.5.10)$$

so this matrix has eigenvalues $\pm 2x_0$ and 0, with the last eigenvalue corresponding to the goldstino mode. In Chapter 29 we will show without the use of perturbation theory that the spontaneous breakdown of supersymmetry requires the existence of goldstinos, and explore their general properties.

26.6 Superspace Integrals, Field Equations, and the Current Superfield

The ‘ $\mathcal{F}$ -terms’ and ‘ D -terms’ from which we construct the Lagrangian density may be expressed as integrals over the superspace coordinates θ_α . The rules for integrals over fermionic parameters originally given by Berezin [4] are derived in Section 9.5. Briefly, because the square of any fermionic parameter vanishes, any function of a set of N fermionic parameters ξ_n may be expressed as

$$f(\xi) = \left(\prod_{n=1}^N \xi_n \right) c + \text{terms with fewer } \xi \text{ factors}, \quad (26.6.1)$$

and its integral over the ξ s is defined simply by

$$\int d^N \xi f(\xi) \equiv c. \quad (26.6.2)$$

The coefficient c may itself depend on other unintegrated c-number variables that anticommute with the ξ s over which we integrate, in which case it is important to standardize the definition of c by moving all ξ s to the left of c before integrating over them, as we have

done in Eq. (26.6.1). With this definition, integration over fermionic variables is a linear operation. It resembles the integral over a real variable in the sense that, since a shift $\xi_n \rightarrow \xi_n + a_n$ of the variable ξ_n by a constant a_n changes the product $\prod_n \xi_n$ only by terms involving fewer ξ factors, it does not affect the value of the integral

$$\int d^N \xi f(\xi + a) = \int d^N \xi f(\xi). \quad (26.6.3)$$

Also, as a special case of Eq. (26.6.2), the integral over N fermionic parameters of a polynomial of order $< N$ vanishes. Integrals over fermionic and bosonic parameters are strikingly different in the way they respond to change of variables: for bosonic parameters x_n we have $d^N x' = \text{Det}(\partial x' / \partial x) d^N x$, while for fermionic parameters

$$d^N \xi' = \left[\text{Det} \left(\frac{\partial \xi'}{\partial \xi} \right) \right]^{-1} d^N \xi. \quad (26.6.4)$$

In particular, the dimensionality of $d\xi$ is *opposite* to the dimensionality of ξ .

According to Eq. (26.2.10), the D -term of a general superfield $S(x, \theta)$ (which may be elementary or composite) is equal, up to a derivative term, to the coefficient of $-(\bar{\theta}\gamma_5\theta)^2/4 = -(\theta^T \epsilon \theta)^2/4$. Any one of the four θ s can be θ_1 , and each possibility gives an equal contribution, so we may assume that θ_1 is the left-most, and pick up a factor of 4. Then θ_2 must be the next-to-left-most. Any of the two remaining θ s may be θ_3 , and each possibility gives an equal contribution, so we may assume that θ_3 is third from the left and pick up a factor of 2, and θ_4 must then be the right-most. That is,

$$-\frac{1}{4}(\bar{\theta}\gamma_5\theta)^2 = -\frac{1}{4} \cdot 4 \cdot 2 \cdot \theta_1\theta_2\theta_3\theta_4,$$

so the coefficient of this function of θ is $-1/2$ the integral over $d^4\theta$. Since this is the D -term up to a derivative, we have then

$$\int d^4x [S]_D = -\frac{1}{2} \int d^4x \int d^4\theta S(x, \theta). \quad (26.6.5)$$

In the same way, using Eq. (26.3.11), we find that the spacetime integral of the $\mathcal{F}$ -term of a general left-chiral superfield Φ (again, either elementary or composite) may be expressed as

$$\int d^4x [\Phi]_{\mathcal{F}} = \frac{1}{2} \int d^4x \int d^2\theta_L \Phi(x, \theta). \quad (26.6.6)$$

Since we are now integrating over θ s, it is convenient to introduce a delta function, defined as usual by the condition that for an arbitrary function $f(\theta)$

$$\int d^4\theta' \delta^4(\theta' - \theta) f(\theta') = f(\theta). \quad (26.6.7)$$

According to Eq. (??), this condition is satisfied by

$$\begin{aligned} \delta^4(\theta' - \theta) &= (\theta'_1 - \theta_1)(\theta'_2 - \theta_2)(\theta'_3 - \theta_3)(\theta'_4 - \theta_4) \\ &= \frac{1}{4} [(\theta_L - \theta'_L)^T \epsilon (\theta_L - \theta'_L)] [(\theta_R - \theta'_R)^T \epsilon (\theta_R - \theta'_R)]. \end{aligned} \quad (26.6.8)$$

The representation of the action as an integral over superspace allows an easy derivation of the field equations in superfield form. Consider, for instance, the action for a set of left-chiral scalar superfields Φ_n (which includes as a special case the general renormalizable theory of left-chiral superfields Φ_n):

$$I = \frac{1}{2} \int d^4x [K(\Phi, \Phi^*)]_D + 2 \operatorname{Re} \int d^4x [f(\Phi)]_{\mathcal{F}}, \quad (26.6.9)$$

with K an arbitrary function of Φ_n and Φ_n^* without derivatives, and f an arbitrary function of Φ_n , also without derivatives. (The motivation for this form of the action and the expression of the action in terms of component fields is described in Section 26.8.) We cannot derive correct field equations by simply demanding that this is stationary with respect to arbitrary variations in Φ , because the Φ_n are constrained by the requirement that they satisfy the requirement $\mathcal{D}_R \Phi_n = 0$ for left-chiral superfields. To make sure that this condition is preserved by any variation, we can use a trick that will also turn out to be useful in deriving superspace Feynman rules in Chapter 30. We write the Φ_n in terms of *potential superfields* $S_n(x, \theta)$ as

$$\Phi_n = \mathcal{D}_R^2 S_n, \quad (26.6.10)$$

from which it follows (using Eq. (26.A.21)) that

$$\Phi_n^* = -\mathcal{D}_L^2 S_n^*, \quad (26.6.11)$$

where $\mathcal{D}_R^2$ and $\mathcal{D}_L^2$ are abbreviations for $(\mathcal{D}_R^T \epsilon \mathcal{D}_R) = -(\bar{\mathcal{D}}_R \mathcal{D}_R)$ and $(\mathcal{D}_L^T \epsilon \mathcal{D}_L) = (\bar{\mathcal{D}}_L \mathcal{D}_L)$, respectively. To see that it is always possible to find an S_n (not necessarily local) that satisfies Eq. (26.6.10), note that for any left-chiral superfields Φ_n ,

$$\mathcal{D}_R^2 \mathcal{D}_L^2 \Phi_n = -16 \square \Phi_n, \quad (26.6.12)$$

so that Eq. (26.6.10) is satisfied by the solution of

$$-16 \square S_n = \mathcal{D}_L^2 \Phi_n. \quad (26.6.13)$$

The expression $\mathcal{D}_R^2 S$ is left-chiral for any S , so the action must be stationary with respect to arbitrary variations in the S_n . Using Eq. (26.6.5), the action may be expressed in terms of the S_n and S_n^* as

$$I = -\frac{1}{4} \int d^4x \int d^4\theta K(-\mathcal{D}_L^2 S^*, \mathcal{D}_R^2 S) + 2 \operatorname{Re} \int d^4x [f(\mathcal{D}_R^2 S)]_{\mathcal{F}}. \quad (26.6.14)$$

The variation of the first term under infinitesimal changes δS_n in the S_n (but not the S_n^*) is easily calculated by an integration by parts in superspace:

$$\begin{aligned} & -\delta \frac{1}{4} \int d^4x \int d^4\theta K(-\mathcal{D}_L^2 S^*, \mathcal{D}_R^2 S) \\ &= -\sum_n \int d^4\theta \delta S_n \mathcal{D}_R^2 \frac{\delta K(-\mathcal{D}_L^2 S^*, \mathcal{D}_R^2 S)}{\delta \mathcal{D}_R^2 S_n}. \end{aligned}$$

Eqs. (26.3.31) and (26.6.5) allow us to express the variation of the integral in the superpotential term under infinitesimal changes δS_n in the S_n as

$$\begin{aligned}\delta \int d^4x [f(\mathcal{D}_R^2 S)]_{\mathcal{F}} &= \sum_n \int d^4x \left[\frac{\partial f(\Phi)}{\partial \Phi_n} \Big|_{\Phi=\mathcal{D}_R^2 S} \mathcal{D}_R^2 \delta S_n \right]_{\mathcal{F}} \\ &= \sum_n \int d^4x \left[\mathcal{D}_R^2 \left(\frac{\partial f(\Phi)}{\partial \Phi_n} \Big|_{\Phi=\mathcal{D}_R^2 S} \delta S_n \right) \right]_{\mathcal{F}} \\ &= 2 \sum_n \int d^4x \left[\frac{\partial f(\Phi)}{\partial \Phi_n} \Big|_{\Phi=\mathcal{D}_R^2 S} \delta S_n \right]_D \\ &= - \sum_n \int d^4x \int d^4\theta \frac{\partial f(\Phi)}{\partial \Phi_n} \Big|_{\Phi=\mathcal{D}_R^2 S} \delta S_n.\end{aligned}$$

The condition for Eq. (26.6.14) to be stationary with respect to arbitrary variations in S_n is then

$$\mathcal{D}_R^2 \frac{\delta K(-\mathcal{D}_L^2 S^*, \mathcal{D}_R^2 S)}{\delta \mathcal{D}_R^2 S_n} = -4 \frac{\partial f(\Phi)}{\partial \Phi_n} \Big|_{\Phi=\mathcal{D}_R^2 S},$$

or in terms of the chiral superfields

$$\mathcal{D}_R^2 \frac{\delta K(\Phi, \Phi^*)}{\delta \Phi_n} = -4 \frac{\partial f(\Phi)}{\partial \Phi_n}. \quad (26.6.15)$$

The complex conjugate yields

$$\mathcal{D}_L^2 \frac{\delta K(\Phi, \Phi^*)}{\delta \Phi_n^*} = 4 \left(\frac{\partial f(\Phi)}{\partial \Phi_n} \right)^*. \quad (26.6.16)$$

It can readily be checked that the components of these equations yield the field equations for the components of Φ_n^* and Φ_n . For instance, recalling that $\mathcal{D}_R^2(\theta_R^T \epsilon \theta_R) = -4$, the θ -independent part of $\mathcal{D}_R^2 \Phi_n^*$ is $4\mathcal{F}_n^*$, while the θ -independent part of $\partial f(\Phi)/\partial \Phi_n$ is $\partial f(\phi)/\partial \phi_n$, so the θ -independent part of Eq. (26.6.15) for $K = \sum_n \Phi_n^* \Phi_n$ yields the relation $\mathcal{F}_n^* = -\partial f(\phi)/\partial \phi_n$, in agreement with Eq. (26.4.6).

As an example of the use of this formalism, let us consider the superfields to which conserved currents belong. Suppose that the superpotential and Kahler potential in the action (26.6.9) are invariant under an infinitesimal global transformation

$$\delta \Phi_n = i\epsilon \sum_m \mathcal{T}_{nm} \Phi_m, \quad \delta \Phi_n^* = -i\epsilon \sum_m \mathcal{T}_{mn} \Phi_m^*, \quad (26.6.17)$$

with ϵ a real infinitesimal constant and $\mathcal{T}_{nm}$ a Hermitian matrix, perhaps part of a Lie algebra of similar transformation matrices. Since the superpotential depends only on the Φ_n , it is automatically also invariant under the extended transformations

$$\delta \Phi_n = i\epsilon \Lambda \sum_m \mathcal{T}_{nm} \Phi_m, \quad \delta \Phi_n^* = -i\epsilon \Lambda^* \sum_m \mathcal{T}_{mn} \Phi_m^*, \quad (26.6.18)$$

where $\Lambda(x, \theta)$ is a superfield that must be taken left-chiral in order that the $\delta \Phi_n$ be left-chiral, but is otherwise unrestricted. On the other hand, other terms such as the Kahler

potential are not in general invariant under these transformations, because $\Lambda \neq \Lambda^*$. For general fields the change in the action must therefore be of the form

$$\delta I = i\epsilon \int d^4x \int d^4\theta [\Lambda - \Lambda^*] \mathcal{J}, \quad (26.6.19)$$

where $\mathcal{J}(x, \theta)$ is some real superfield, known as the *current superfield*. But if the field equations are satisfied then the action is stationary under *any* variation in the superfields, so the integral (26.6.19) must vanish for any left-chiral superfield $\Lambda(x, \theta)$. Any such Λ may be put in the form $\Lambda = \mathcal{D}_R^2 S$, so this means that the current superfield must satisfy

$$\mathcal{D}_R^2 \mathcal{J} = \mathcal{D}_L^2 \mathcal{J} = 0. \quad (26.6.20)$$

That is, $\mathcal{J}$ is a *linear* superfield. As we saw in Section 26.3, this means that its components satisfy

$$N^{\mathcal{J}} = M^{\mathcal{J}} = \partial^\mu V_\mu^{\mathcal{J}} = 0, \quad \lambda^{\mathcal{J}} = -\not{\partial} \omega^{\mathcal{J}}, \quad D^{\mathcal{J}} = -\square C^{\mathcal{J}}. \quad (26.6.21)$$

This allows us to identify the V -component $V_\mu^{\mathcal{J}}$ as the conserved current associated with this symmetry.

For the particular action (26.6.9), the current superfield takes the form

$$\mathcal{J} = \sum_{nm} \frac{\partial K(\Phi, \Phi^*)}{\partial \Phi_n} \mathcal{T}_{nm} \Phi_m = \sum_{nm} \frac{\partial K(\Phi, \Phi^*)}{\partial \Phi_n^*} \mathcal{T}_{mn} \Phi_m^*, \quad (26.6.22)$$

with the equality of these two expressions a consequence of symmetry under the transformation (26.6.17). Then, using the field equation (26.6.15),

$$\mathcal{D}_R^2 \mathcal{J} = \sum_{nm} \left[\mathcal{D}_R^2 \frac{\partial K(\Phi, \Phi^*)}{\partial \Phi_n} \right] \mathcal{T}_{nm} \Phi_m = -4 \sum_{nm} \frac{\partial f(\Phi)}{\partial \Phi_n} \mathcal{T}_{nm} \Phi_m, \quad (26.6.23)$$

which vanishes because of the assumed invariance of the superpotential under the transformation (26.6.17). In the same way, using the second of the two expressions for $\mathcal{J}$ and the field equation (26.6.16), we find that $\mathcal{D}_L^2 \mathcal{J} = 0$, thus verifying the conservation condition (26.6.20).

26.7 The Supercurrent

Like any other continuous global symmetry, supersymmetry leads to the existence of a conserved current [5]. The conservation and commutation properties of the supersymmetry current are operator equations that will remain valid even when supersymmetry is spontaneously broken, and that will therefore be useful to us in Chapter 29, when we consider theories of spontaneous supersymmetry breaking in a non-perturbative context. Also, the supersymmetry current is related to components of a superfield known as the *supercurrent* [6], that will be of fundamental importance in our treatment of supergravity in Chapter 31.

As we saw in Section 7.3, the existence of an ordinary global symmetry of the Lagrangian density under an infinitesimal transformation $\chi^\ell \longrightarrow \chi^\ell + \epsilon \mathcal{F}^\ell$ (with χ^ℓ a generic

canonical or auxiliary boson or fermion field and $\mathcal{F}^\ell$ a function of the canonical and auxiliary fields) leads to the existence of a current

$$J^\mu(x) \propto \sum_\ell \frac{\partial \mathcal{L}(x)}{\partial (\partial_\mu \chi^\ell(x))} \mathcal{F}^\ell(x),$$

which is conserved for fields satisfying the field equations, and generates the symmetry in the sense that the canonical commutation relations give

$$\left[\int d^3x J^0(x), \chi^\ell(y) \right] \propto \mathcal{F}^\ell(y).$$

The supersymmetry current requires a somewhat more complicated treatment for two reasons. One is that supersymmetry is only a symmetry of the action, not of the Lagrangian density or the Lagrangian. Instead, the variation of the Lagrangian density under an infinitesimal supersymmetry transformation is a spacetime derivative, which we may write in the form

$$\delta \mathcal{L} = \sum_\ell (\bar{\alpha} \partial_\mu K^\mu), \quad (26.7.1)$$

with K^μ a four-vector of Majorana spinors. In consequence, the supersymmetry current is not the usual Noether current. The Noether current is a four-vector N^μ of Majorana spinors defined by

$$\sum_\ell \frac{\partial_R \mathcal{L}}{\partial (\partial_\mu \chi^\ell)} \delta \chi^\ell \equiv -(\bar{\alpha} N^\mu), \quad (26.7.2)$$

whose divergence is given by the Euler–Lagrange equations as

$$\begin{aligned} (\bar{\alpha} \partial_\mu N^\mu) &= \sum_\ell \frac{\partial_R \mathcal{L}}{\partial \chi^\ell} \delta \chi^\ell - \sum_\ell \frac{\partial_R \mathcal{L}}{\partial (\partial_\mu \chi^\ell)} \partial_\mu \delta \chi^\ell \\ &= -\delta \mathcal{L}. \end{aligned} \quad (26.7.3)$$

(Here ∂_R denotes the right partial derivative, defined by moving differentiated fermionic variables to the right before differentiating.) Instead, we must define the supersymmetry current as

$$S^\mu \equiv N^\mu + K^\mu, \quad (26.7.4)$$

which Eqs. (26.7.1) and (26.7.3) tell us is conserved:

$$\partial_\mu S^\mu = 0. \quad (26.7.5)$$

The second complication is that the change $\delta \chi^\ell$ in a canonical field χ^ℓ under a supersymmetry transformation is not just a function of canonical fields, but also involves their canonical conjugates. For instance, Eq. (26.3.15) shows that the change in the ψ -component of a chiral scalar superfield involves the time derivative of the ϕ -component. As a result the commutator of the Noether charge $\int d^3x N^0$ with a general canonical field does not give the supersymmetry transformation of that field. Fortunately, this complication is cancelled by the first complication: when the commutators of $\int d^3x K^0$ as well as $\int d^3x N^0$

with the fields are taken into account, the operator $\int d^3x S^0$ does generate supersymmetry transformations⁵ in the sense that

$$\left[\int d^3x (\bar{\alpha} S^0), \chi^\ell \right] = i\delta\chi^\ell, \quad (26.7.6)$$

consistently with Eqs. (26.2.1) and (26.2.8).

For example, we may derive an explicit formula for the supersymmetry current in the general renormalizable theory of left-chiral superfields Φ_n , which can be used to check that it does generate supersymmetry transformations, in the sense of Eq. (26.7.6). The Lagrangian density (26.4.7) for this theory may be put in the form

$$\mathcal{L} = \sum_n \left[-\partial_\mu \phi_n^* \partial^\mu \phi_n - \frac{1}{2} (\overline{\psi_{Ln}} \gamma^\mu \partial_\mu \psi_{Ln}) - \frac{1}{2} (\overline{\psi_{Rn}} \gamma^\mu \partial_\mu \psi_{Rn}) \right] + \text{non-derivative terms}. \quad (26.7.7)$$

Using the transformation rules (26.3.15), (26.3.17), (26.3.18), and (26.3.20) (with $\tilde{\phi} = \phi^*$),

⁵This is a general result for currents constructed in this way. For instance, consider a Lagrangian L (not Lagrangian density) that depends on a set of canonical variables q^n and their time derivatives $\dot{q}^n$, with no constraints of any class. In quantum field theory the label n includes spatial coordinates as well as discrete spin and species labels, and $L = \int d^3x \mathcal{L}$. Our assumption here that the Lagrangian density is invariant up to spacetime derivatives under some infinitesimal transformation δ means that δL is the time derivative of some functional F . That is,

$$\sum_n \frac{\partial L}{\partial q^n} \delta q^n + \sum_n \frac{\partial L}{\partial \dot{q}^n} \delta \dot{q}^n = \frac{d}{dt} F.$$

Using the canonical equations of motion, this can be written as a conservation law $\dot{Q} = 0$, where the conserved charge is

$$Q = - \sum_n \frac{\partial L}{\partial \dot{q}^n} \delta q^n + F.$$

In our case here, $Q = \int d^3x [N^0 + K^0]$. We assume the usual unconstrained commutation relations

$$\left[\frac{\partial L}{\partial \dot{q}^n}, q^m \right] = -i\delta_n^m, \quad [q^n, q^m] = 0,$$

and find the commutator

$$[Q, q^m] = i\delta q^m - \sum_{n\ell} \frac{\partial L}{\partial \dot{q}^\ell} \frac{\partial \delta q^\ell}{\partial \dot{q}^n} [\dot{q}^n, q^m] + \sum_n \frac{\partial F}{\partial \dot{q}^n} [\dot{q}^n, q^m].$$

To evaluate the second and third terms, we note that the second time derivatives $\ddot{q}^n$ appear linearly in the invariance condition, so their coefficients must match: even without using the equations of motion, we have

$$\sum_\ell \frac{\partial L}{\partial \dot{q}^\ell} \frac{\partial \delta q^\ell}{\partial \dot{q}^n} = \frac{\partial F}{\partial \dot{q}^n}.$$

The second and third terms in the commutator therefore cancel, leaving us with the desired result

$$[Q, q^m] = i\delta q^m.$$

Taking the time derivative gives also

$$[Q, \dot{q}^m] = i\delta \dot{q}^m.$$

This result has been extended to theories with constraints [7].

the Noether current defined by Eq. (26.7.2) is

$$N^\mu = \frac{1}{\sqrt{2}} \sum_n \left[2(\partial^\mu \phi_n^*) \psi_{nL} + 2(\partial^\mu \phi_n) \psi_{nR} + (\not{\partial} \phi_n) \gamma^\mu \psi_{nR} + (\not{\partial} \phi_n^*) \gamma^\mu \psi_{nL} \right. \\ \left. - \mathcal{F}_n \gamma^\mu \psi_{nR} - \mathcal{F}_n^* \gamma^\mu \psi_{nL} \right]. \quad (26.7.8)$$

We can calculate the change in the Lagrangian density either directly, or, more easily, by noting that the changes of D -terms and $\mathcal{F}$ -terms under supersymmetry transformations are given respectively by Eqs. (26.2.17) and (26.3.16). Either way, we find that the current K^μ in Eq. (26.7.1) is

$$K^\mu = \frac{1}{\sqrt{2}} \sum_n \gamma^\mu \left[-(\not{\partial} \phi_n) \psi_{nR} - (\not{\partial} \phi_n^*) \psi_{nL} + \mathcal{F}_n^* \psi_{nL} + \mathcal{F}_n \psi_{nR} \right. \\ \left. + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right) \psi_{nL} + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^* \psi_{nR} \right]. \quad (26.7.9)$$

Adding (26.7.8) and (26.7.9) gives the supersymmetry current for this class of theories

$$S^\mu = \sqrt{2} \sum_n \left[(\not{\partial} \phi_n) \gamma^\mu \psi_{nR} + (\not{\partial} \phi_n^*) \gamma^\mu \psi_{nL} \right. \\ \left. + \left(\frac{\partial f}{\partial \phi_n} \right) \gamma^\mu \psi_{nL} + \left(\frac{\partial f}{\partial \phi_n} \right)^* \gamma^\mu \psi_{nR} \right]. \quad (26.7.10)$$

It is straightforward then to use the canonical commutation and anticommutation relations to verify that $\int d^3x S^0$ satisfies the commutation relations (26.7.6).

There is another definition of symmetry currents, in terms of the response of the matter action to a local symmetry transformation, which is particularly useful when the associated symmetries are ‘gauged,’ as supersymmetry will be when we turn to supergravity theories in Chapter 31. In the absence of supergravity fields, the action is not invariant under local supersymmetry transformations. If we make such a transformation with a spacetime-dependent parameter $\alpha(x)$, the action will change by an amount that, in order to vanish when $\alpha(x)$ is constant, must (even when the field equations are not satisfied) be of the form

$$\delta I = - \int d^4x \left((\partial_\mu \bar{\alpha}(x)) S^\mu(x) \right), \quad (26.7.11)$$

with $S^\mu(x)$ a four-vector of Majorana spinor operator coefficients. This does not define $S^\mu(x)$ uniquely, because when we generalize global supersymmetry transformations to local transformations, we might in general give the change $\delta\chi$ of a field χ under a local supersymmetry transformation an arbitrary dependence on the derivatives of $\alpha(x)$. There is, however, one way of defining local supersymmetry transformations that guarantees that the coefficient $S^\mu(x)$ in Eq. (26.7.11) is the same as the current defined by Eq. (26.7.4), which as we have seen generates the symmetry transformations in the sense of Eq. (26.7.6). It is to specify that *derivatives of $\alpha(x)$ do not appear in the supersymmetry transformation of the canonical or auxiliary fields χ^ℓ* . For instance, the local versions of the transformation rules (26.3.15)–(26.3.17) for the components of a left-chiral superfield are

$$\delta\psi_L(x) = \sqrt{2} \partial_\mu \phi(x) \gamma^\mu \alpha_R(x) \phi(x) + \sqrt{2} \mathcal{F}(x) \alpha_L(x), \quad (26.7.12)$$

$$\delta\mathcal{F}(x) = \sqrt{2}(\bar{\alpha}_L(x)\not{\partial}\psi_L(x)) , \quad (26.7.13)$$

$$\delta\phi(x) = \sqrt{2}(\bar{\alpha}_R(x)\psi_L(x)) . \quad (26.7.14)$$

Eq. (26.3.21) shows that the superfield may be expressed in terms of its component fields at x_+^μ without derivatives, so the transformation rule for the superfield may be expressed as

$$\delta\Phi(x, \theta) = (\bar{\alpha}(x_+)\mathcal{Q})\Phi(x, \theta) , \quad (26.7.15)$$

where $\mathcal{Q}$ is the operator (26.2.2).

With local supersymmetry transformations defined in this way, the change of the action that they induce consists of two terms. First, although the variation in canonical fields under a supersymmetry transformation does not involve derivatives of $\alpha(x)$, the variation in derivatives of canonical fields does. This produces a change in the Lagrangian density which is the same as Eq. (26.7.2), except that $\bar{\alpha}$ is replaced with $\partial_\mu\bar{\alpha}$:

$$\delta_1 I = - \int d^4x ([\partial_\mu\bar{\alpha}(x)]N^\mu(x)) .$$

The second term in the change of the action arises from the fact that the Lagrangian density is not invariant even under the part of the supersymmetry transformation that does not involve derivatives of $\alpha(x)$. According to Eq. (26.7.1), this produces a change in the action

$$\delta_2 I = \int d^4x (\bar{\alpha}(x)\partial_\mu K^\mu(x)) = - \int d^4x ((\partial_\mu\bar{\alpha}(x))K^\mu(x)) .$$

Adding $\delta_1 I$ and $\delta_2 I$ gives a total change in the action of the form (26.7.11), with $S^\mu(x)$ given by Eq. (26.7.4), as was to be shown.

Even with this specification of the transformation property of the component fields, the supersymmetry current $S^\mu(x)$ is not uniquely specified by Eq. (26.7.11), because we could always introduce a modified current

$$S_{\text{new}}^\mu = S^\mu + \partial_\nu A^{\mu\nu} , \quad (26.7.16)$$

where $A^{\mu\nu} = -A^{\nu\mu}$ is an arbitrary antisymmetric tensor of Majorana spinors. The term $\partial_\nu A^{\mu\nu}$ is conserved whether or not the field equations are satisfied, and its time component is a space derivative, so $\int d^3x S_{\text{new}}^0 = \int d^3x S^0$, leaving Eq. (26.7.6) unchanged.

There is in fact a particular choice of $A^{\mu\nu}$ with the convenient feature that $\gamma_\mu S_{\text{new}}^\mu$ turns out to be a measure of the violation of scale invariance by the theory. By using the Dirac equations derived from the Lagrangian density (26.4.7):

$$\not{\partial}\psi_{mL} = - \sum_n \left(\frac{\partial^2 f(\phi)}{\partial\phi_m \partial\phi_n} \right)^* \psi_{nR} , \quad \not{\partial}\psi_{mR} = - \sum_n \left(\frac{\partial^2 f(\phi)}{\partial\phi_m \partial\phi_n} \right) \psi_{nL} , \quad (26.7.17)$$

it is straightforward to calculate that

$$\begin{aligned} \gamma_\mu S^\mu = -2\sqrt{2} \sum_n \left\{ \not{\partial} (\phi_n \psi_{nR} + \phi_n^* \psi_{nL}) \right. \\ \left. + \left(\sum_m \phi_m \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} - 2 \frac{\partial f(\phi)}{\partial \phi_n} \right) \psi_{nL} \right. \\ \left. + \left(\sum_m \phi_m \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} - 2 \frac{\partial f(\phi)}{\partial \phi_n} \right)^* \psi_{nR} \right\}. \end{aligned}$$

We can eliminate the first term by introducing a modified supersymmetry current of the general type of Eq. (26.7.16):

$$S_{\text{new}}^\mu = S^\mu + \frac{\sqrt{2}}{3} [\gamma^\mu, \gamma^\nu] \sum_n \partial_\nu (\phi_n \psi_{nR} + \phi_n^* \psi_{nL}), \quad (26.7.18)$$

for which

$$\begin{aligned} \gamma_\mu S_{\text{new}}^\mu = -2\sqrt{2} \sum_n \left\{ \left(\sum_m \phi_m \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} - 2 \frac{\partial f(\phi)}{\partial \phi_n} \right) \psi_{nL} \right. \\ \left. + \left(\sum_m \phi_m \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} - 2 \frac{\partial f(\phi)}{\partial \phi_n} \right)^* \psi_{nR} \right\}. \end{aligned} \quad (26.7.19)$$

The right-hand side vanishes for a scale-invariant Lagrangian density, with $f(\Phi)$ a homogeneous polynomial of third order in the Φ_n .

We now turn to the supersymmetry transformation property of the supersymmetry current. It is straightforward to check that the current given by Eqs. (26.7.18) and (26.7.10) is related to the ω -component ω_μ^Θ of a real non-chiral superfield Θ_μ by⁶

$$S_{\text{new}}^\mu = -2\omega^\Theta{}^\mu + 2\gamma^\mu \gamma^\nu \omega_\nu^\Theta, \quad (26.7.20)$$

where

$$\Theta_\mu = \frac{i}{12} \sum_n [4\Phi_n^* \partial_\mu \Phi_n - 4\Phi_n \partial_\mu \Phi_n^* + ((\bar{D}\Phi_n^*) \gamma_\mu (D\Phi_n))]. \quad (26.7.21)$$

The superfield Θ^μ is known as the *supercurrent*.

The supercurrent obeys a conservation law that incorporates the conservation of the supersymmetry current (26.7.20) and much else besides. To derive it, we may use the

⁶Here we introduce a notation that will be used extensively in Chapter 31; following Eq. (26.2.10), the components C^S , ω^S , M^S , N^S , V_ν^S , λ^S , and D^S of an arbitrary superfield $S(x, \theta)$ are defined by the expansion

$$\begin{aligned} S(x, \theta) = C^S(x) - i(\bar{\theta} \gamma_5 \omega^S(x)) - \frac{i}{2} (\bar{\theta} \gamma_5 \theta) M^S(x) - \frac{1}{2} (\bar{\theta} \theta) N^S(x) \\ + \frac{i}{2} (\bar{\theta} \gamma_5 \gamma^\nu \theta) V_\nu^S(x) - i(\bar{\theta} \gamma_5 \theta) \left(\bar{\theta} \left[\lambda^S(x) + \frac{1}{2} \not{\partial} \omega^S(x) \right] \right) \\ - \frac{1}{4} (\bar{\theta} \gamma_5 \theta)^2 \left[D^S(x) + \frac{1}{2} \square C^S(x) \right]. \end{aligned}$$

anticommutation relation (26.2.30) to write⁷

$$[\mathcal{D}_R, (\bar{\mathcal{D}}_L \mathcal{D}_L)] = -4 \not{\mathcal{D}} \mathcal{D}_L.$$

Together with the chirality conditions $\mathcal{D}_R \Phi_n = \mathcal{D}_L \Phi_n^* = 0$, this gives

$$\begin{aligned} \gamma^\mu \mathcal{D}_L \sum_n [\Phi_n^* \partial_\mu \Phi_n - \Phi_n \partial_\mu \Phi_n^*] &= -\frac{1}{4} \sum_n \Phi_n^* \mathcal{D}_R (\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n \\ &\quad - \sum_n (\not{\mathcal{D}} \Phi_n^*) \mathcal{D}_L \Phi_n \end{aligned}$$

and

$$\gamma^\mu \mathcal{D}_L \sum_n ((\bar{\mathcal{D}} \Phi_n^*) \gamma_\mu (\mathcal{D} \Phi_n)) = 4 \sum_n (\not{\mathcal{D}} \Phi_n^*) \mathcal{D} \Phi_n + 2 \sum_n \mathcal{D} \Phi_n^* (\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n,$$

so that the superfield (26.7.21) satisfies

$$\begin{aligned} \gamma^\mu \mathcal{D}_L \Theta^\mu &= \frac{i}{6} \sum_n (\mathcal{D}_R \Phi_n^*) (\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n \\ &\quad - \frac{i}{12} \sum_n \Phi_n^* \mathcal{D}_R (\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n. \end{aligned} \tag{26.7.22}$$

We saw in Section 26.6 that the field equations for the Lagrangian density (26.4.7) may be expressed in the form

$$(\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n = -4 \left(\frac{\partial f(\Phi)}{\partial \Phi_n} \right)^*. \tag{26.7.23}$$

Using this in Eq. (26.7.22) gives finally

$$\begin{aligned} \gamma^\mu \mathcal{D}_L \Theta_\mu &= -\frac{2i}{3} \sum_n (\mathcal{D}_R \Phi_n^*) \left(\frac{\partial f}{\partial \Phi_n} \right)^* + \frac{i}{3} \sum_n \Phi_n^* \mathcal{D}_R \left(\frac{\partial f}{\partial \Phi_n} \right)^* \\ &= \frac{i}{3} \mathcal{D}_R \left[\sum_n \Phi_n \frac{\partial f(\Phi)}{\partial \Phi_n} - 3f(\Phi) \right]^*. \end{aligned} \tag{26.7.24}$$

The Hermitian adjoint of Eq. (26.7.24) is

$$\gamma^\mu \mathcal{D}_R \Theta_\mu = -\frac{i}{3} \mathcal{D}_L \left[\sum_n \Phi_n \frac{\partial f(\Phi)}{\partial \Phi_n} - 3f(\Phi) \right]. \tag{26.7.25}$$

The sum of this and Eq. (26.7.24) then gives the conservation law

$$\gamma^\mu \mathcal{D} \Theta_\mu = \mathcal{D} X, \tag{26.7.26}$$

where X is a real chiral superfield given (up to an additive constant) in this class of theories by

$$X = \frac{2}{3} \text{Im} \left[\sum_n \Phi_n \frac{\partial f(\Phi)}{\partial \Phi_n} - 3f(\Phi) \right]. \tag{26.7.27}$$

⁷It should be noted that $\bar{\mathcal{D}}_L$ and $\bar{\mathcal{D}}_R$ are the left- and right-handed components of the covariant adjoint $\bar{\mathcal{D}}$, rather than the covariant adjoints $\overline{\mathcal{D}_L}$ and $\overline{\mathcal{D}_R}$ of $\mathcal{D}_L$ and $\mathcal{D}_R$.

Although it has been derived here only for renormalizable theories of chiral superfields, we may expect that the conservation law (26.7.26) holds more generally, though of course with X not necessarily given by Eq. (26.7.27), because of the other conservation laws that it incorporates. (A generalized formula for X will be given in Section 31.4.) To derive these relations, we must use Eq. (26.2.10) to express Θ_μ in terms of components C_μ^Θ , ω_μ^Θ , etc. and use Eq. (26.3.9) to express the chiral superfield X in terms of components A^X , ψ^X , etc. With the aid of Eqs. (26.A.9), (26.A.16), (26.A.17), and the Dirac matrix identities

$$[\gamma^\rho, \gamma^\sigma] = -\frac{i}{2}\epsilon^{\rho\sigma\mu\nu}\gamma_5[\gamma_\mu, \gamma_\nu], \quad (26.7.28)$$

$$\gamma^\mu\gamma^\rho\gamma^\nu = \eta^{\mu\rho}\gamma^\nu - \eta^{\mu\nu}\gamma^\rho + \eta^{\nu\rho}\gamma^\mu + i\gamma_5\epsilon^{\mu\nu\rho\sigma}\gamma_\sigma, \quad (26.7.29)$$

we can then expand both sides of Eq. (26.7.26) in the terms

$$\begin{aligned} &1, \quad \theta, \quad \gamma_5\theta, \quad \gamma^\nu\theta, \quad \gamma_5\gamma^\nu\theta, \quad \gamma_5[\gamma^\mu, \gamma^\nu]\theta, \\ &(\bar{\theta}\theta), \quad (\bar{\theta}\gamma_5\theta), \quad (\bar{\theta}\gamma_5\gamma^\nu\theta), \\ &\theta(\bar{\theta}\gamma_5\theta), \quad \gamma_5\theta(\bar{\theta}\gamma_5\theta), \quad \gamma^\nu\theta(\bar{\theta}\gamma_5\theta), \\ &\gamma^\nu\gamma_5\theta(\bar{\theta}\gamma_5\theta), \quad [\gamma^\rho, \gamma^\sigma]\theta(\bar{\theta}\gamma_5\theta), \quad (\bar{\theta}\gamma_5\theta)^2. \end{aligned}$$

Matching the coefficients of $1, \theta, \gamma_5\theta, \gamma^\nu\theta, \gamma_5\gamma^\nu\theta, \gamma_5[\gamma^\mu, \gamma^\nu]\theta$, respectively, yields the results⁸

$$\psi^X = -i\gamma_5\gamma^\mu\omega_\mu^\Theta, \quad (26.7.30)$$

$$F^X = \partial^\mu C_\mu^\Theta, \quad (26.7.31)$$

$$G^X = (V^\Theta)^\mu{}_\mu, \quad (26.7.32)$$

$$\partial_\mu A^X = -N_\mu^\Theta, \quad (26.7.33)$$

$$\partial_\mu B^X = M_\mu^\Theta, \quad (26.7.34)$$

$$0 = V_{\mu\nu}^\Theta - V_{\nu\mu}^\Theta + \epsilon_{\mu\nu\rho\sigma}\partial^\sigma C^{\Theta\rho}. \quad (26.7.35)$$

Matching the coefficients of either $(\bar{\theta}\theta)$ or $(\bar{\theta}\gamma_5\theta)$ yields the same result:

$$0 = \gamma^\mu\lambda_\mu^\Theta, \quad (26.7.36)$$

and matching the coefficients of $(\bar{\theta}\gamma_5\gamma^\nu\theta)$ yields the result

$$-i\gamma_5[\gamma^\nu, \not{\partial}]\psi^X = 2\gamma^\mu\gamma^\nu\lambda_\mu^\Theta + \gamma^\mu[\gamma^\nu, \not{\partial}]\omega_\mu^\Theta. \quad (26.7.37)$$

From Eqs. (26.7.30), (26.7.36), and (26.7.37) we obtain the conservation of the supersymmetry current (26.7.20):

$$0 = \partial_\mu S_{\text{new}}^\mu = -2\partial^\mu\omega_\mu^\Theta + 2\not{\partial}\gamma^\mu\omega_\mu^\Theta, \quad (26.7.38)$$

and a relation among λ_μ^Θ and ω_μ^Θ :

$$\lambda_\nu^\Theta = -\not{\partial}\omega_\nu^\Theta + \partial_\nu\gamma^\mu\omega_\mu^\Theta. \quad (26.7.39)$$

⁸Note that $V_{\mu\nu}^\Theta$ is the V_ν -component of Θ_μ , not the V_μ -component of Θ_ν .

Matching the coefficients of $\theta(\bar{\theta}\gamma_5\theta)$ and $\gamma_5\theta(\bar{\theta}\gamma_5\theta)$ gives relations that can be obtained by taking the divergence of Eqs. (26.7.34) and (26.7.33), respectively. Matching the coefficients of $\gamma^\rho\theta(\bar{\theta}\gamma_5\theta)$ gives

$$\partial_\rho G^X = \partial^\mu V_{\mu\rho}^\Theta + \partial^\mu V_{\rho\mu}^\Theta - \partial_\rho V^{\Theta\lambda}_\lambda, \quad (26.7.40)$$

which, combined with Eq. (26.7.32), yields the conservation law

$$\partial_\mu T^{\mu\nu} = 0, \quad (26.7.41)$$

where $T^{\mu\nu}$ is the symmetric tensor

$$T_{\mu\nu} \equiv -\frac{1}{2}V_{\mu\nu}^\Theta - \frac{1}{2}V_{\nu\mu}^\Theta + \eta_{\mu\nu}V^{\Theta\lambda}_\lambda. \quad (26.7.42)$$

Matching the coefficients of $\gamma^\rho\gamma_5\theta(\bar{\theta}\gamma_5\theta)$ gives

$$\partial_\mu F^X = 2D_\mu^\Theta + \square C_\mu^\Theta + \epsilon_{\rho\nu\sigma\mu}\partial^\nu V^{\Theta\rho\sigma}, \quad (26.7.43)$$

which, with Eqs. (26.7.31) and (27.6.35), yields a relation between D_μ^Θ and C_μ^Θ :

$$D_\mu^\Theta = -\square C_\mu^\Theta + \partial_\mu\partial^\nu C_\nu^\Theta. \quad (26.7.44)$$

Matching the coefficients of $[\gamma^\rho, \gamma^\sigma]\theta(\bar{\theta}\gamma_5\theta)$ and $(\bar{\theta}\gamma_5\theta)^2$ gives results that already follow from Eq. (26.7.34) and from Eqs. (26.7.38) and (26.7.39), respectively.

The conserved symmetric tensor $T^{\mu\nu}$ may be identified as the energy-momentum tensor of the system. To check this, we use Eqs. (26.1.18) and (26.2.12) to write the change in $\omega_\mu^\Theta(x)$ under a supersymmetry transformation with infinitesimal parameter α as

$$\begin{aligned} \delta\omega_\mu^\Theta &= -i[(\bar{Q}\alpha), \omega_\mu^\Theta] = +i[\omega_\mu^\Theta, (\bar{Q}\alpha)] \\ &= (-i\gamma_5\bar{\theta}C_\mu^\Theta - M_\mu^\Theta + i\gamma_5N_\mu^\Theta + \gamma^\nu V_{\mu\nu}^\Theta)\alpha. \end{aligned}$$

Eqs. (26.7.33)–(26.7.35) allow us to put this in the form

$$\begin{aligned} i\{\omega_\mu^\Theta, \bar{Q}\} &= \frac{1}{2}\gamma^\nu(V_{\mu\nu}^\Theta + V_{\nu\mu}^\Theta) - \partial_\mu(B^X + \gamma_5A^X) - i\gamma_5\bar{\theta}C_\mu^\Theta \\ &\quad + \frac{1}{2}\epsilon_{\mu\nu\kappa\sigma}\gamma^\nu\partial^\kappa C^{\Theta\sigma}. \end{aligned}$$

In terms of the currents (26.7.20) and (26.7.42), this reads

$$\begin{aligned} i\{S_{\text{new}}^\mu, \bar{Q}\} &= 2\gamma_\nu T^{\mu\nu} + 2(\partial^\mu - \gamma^\mu\bar{\theta})(B^X + \gamma_5A^X) - \epsilon^{\mu\nu\kappa\sigma}\gamma_\nu\partial_\kappa C_\sigma^\Theta \\ &\quad + 2i\gamma_5\left(\bar{\theta}C^{\Theta\mu} - \gamma^\mu\gamma^\lambda\bar{\theta}C_\lambda^\Theta - \frac{1}{2}\gamma^\mu[\bar{\theta}, \gamma^\sigma]C_\sigma^\Theta\right). \end{aligned} \quad (26.7.45)$$

For $\mu = 0$ all the terms on the right except the first are space derivatives, and therefore vanish when we integrate over space, leaving us with

$$i\left\{\int d^3x S_{\text{new}}^0, \bar{Q}\right\} = 2\gamma_\nu \int d^3x T^{0\nu}. \quad (26.7.46)$$

We have defined the supersymmetry current S_{new}^μ to give $\int d^3x S_{\text{new}}^0 = Q$, so the fundamental anticommutation relation (??) tells us that

$$\int d^3x T^{0\nu} = P^\nu, \quad (26.7.47)$$

which, with the conservation condition (26.7.41), allows us to identify $T^{\mu\nu}$ as the energy-momentum tensor.

It is important to note *which* energy-momentum tensor we have constructed in this way. Either directly from Eq. (26.7.21), or by considering the supersymmetry transformation of the current (26.7.18), we can calculate that the energy-momentum tensor $T^{\mu\nu}$ for renormalizable theories of chiral superfields is

$$\begin{aligned} T^{\mu\nu} = & \sum_n [\partial^\mu \phi_n^* \partial^\nu \phi_n + \partial^\nu \phi_n^* \partial^\mu \phi_n] \\ & - \eta^{\mu\nu} \sum_n \left[\partial^\lambda \phi_n^* \partial_\lambda \phi_n + \left| \frac{\partial f(\phi)}{\partial \phi_n} \right|^2 \right] \\ & + \frac{1}{3} (\eta^{\mu\nu} \square - \partial^\mu \partial^\nu) \sum_n |\phi_n|^2 + \dots, \end{aligned} \quad (26.7.48)$$

where the dots denote terms involving fermions, which do not concern us here. We see that the last term, which is related by supersymmetry to the correction term in Eq. (26.7.18), has the effect that the energy-momentum tensor is traceless for the massless free-field theory with no superpotential, in which case $\square \phi_n = 0$. A simple calculation shows that $T^{\mu\nu}$ is also traceless more generally for scale-invariant theories, with $f(\phi)$ a homogeneous polynomial of third order in the ϕ_n .

Supersymmetry also imposes an interesting relation among violations of scale invariance and R conservation. Eqs. (26.7.30)–(26.7.32) show that $\gamma_\mu S_{\text{new}}^\mu = 6\gamma_\mu \omega^{\Theta\mu}$, $\partial^\mu C_\mu^\Theta$, and $T^\lambda{}_\lambda = 2V^{\Theta\mu}{}_\mu$ (which measures the violation of scale invariance) are proportional to components of a chiral superfield X , so if any one of these vanishes as an operator equation (that is, not just for some particular field configuration), then they all do. In this case, we can show that $C^{\Theta\rho}$ is proportional to the current of an R quantum number. To see this, note that Eq. (26.2.11) gives

$$\delta C_\sigma^\Theta = i[C_\sigma^\Theta, (\bar{\alpha}Q)] = i(\bar{\alpha}\gamma_5\omega_\sigma^\Theta),$$

so that in general

$$[C_\sigma^\Theta, Q] = \gamma_5\omega_\sigma^\Theta. \quad (26.7.49)$$

We have seen that if C_σ^Θ is conserved then $\gamma_\mu S^\mu = 0$, so that Eq. (26.7.20) gives $S_\sigma = -2\omega_\sigma^\Theta$. Setting $\sigma = 0$ and integrating over $\mathbf{x}$ in Eq. (26.7.49) then gives

$$\left[\int d^3x C^{0\Theta}, Q \right] = -\frac{1}{2}\gamma_5 Q. \quad (26.7.50)$$

We can therefore introduce a current

$$\mathcal{R}^\mu \equiv 2C^{\mu\Theta}, \quad (26.7.51)$$

which *if* conserved is the current of a quantum number $\mathcal{R} \equiv \int d^3x \mathcal{R}^0$ for which Q_L and Q_R destroy the values $+1$ and -1 , respectively. Since the commutator of Q_L with a scalar superfield Φ involves a term $\partial\Phi/\partial\theta_L$, this means that θ_L carries the $\mathcal{R}$ value $+1$, in accord with the usual definition. A theory in which the superfield X vanishes, or, equivalently, in which $T^\mu{}_\mu$, $\gamma_\mu S^\mu$, and $\partial_\mu \mathcal{R}^\mu$ all vanish, is invariant under an enlarged set of supersymmetry transformations, generated by the superconformal algebra described at the end of Section 25.2.

In scale-invariant theories the value of the $\mathcal{R}$ quantum number carried by various superfields is fixed by the structure of the Lagrangian. For instance, in a scale-invariant theory of chiral scalar superfields, the superpotential must be a homogeneous polynomial of third order in the superfields. The $\mathcal{F}$ -term of the superpotential is proportional to the coefficient of θ_L^2 , which has $\mathcal{R}$ quantum number $+2$, so the $\mathcal{R}$ quantum number of the $\mathcal{F}$ -term of the superpotential is the $\mathcal{R}$ quantum number of the superpotential itself minus two. $\mathcal{R}$ invariance then requires that we give the scalar superfields an $\mathcal{R}$ quantum number $+2/3$, so that the superpotential will have $\mathcal{R}$ quantum number $+2$, and its $\mathcal{F}$ -term will have $\mathcal{R}$ quantum number zero. That is, the scalar components ϕ_n have $\mathcal{R} = 2/3$ and the spinor components ψ_{nL} (proportional to the coefficient of θ_L in the superfield) have $\mathcal{R} = -1/3$. This can be verified by calculating the current $\mathcal{R}^\mu$ from the C -term of the supercurrent (26.7.21) for this class of theories:

$$\mathcal{R}_\mu = \frac{2i}{3}[\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*] - \frac{i}{6}(\bar{\psi} \gamma_\mu \gamma_5 \psi). \quad (26.7.52)$$

(The second term contains an extra factor of $1/2$ because ψ is a Majorana spinor.)

Quantum corrections can introduce violations of $\mathcal{R}$ invariance (through Adler–Bell–Jackiw anomalies) and of scale invariance (through the renormalization group running of the coupling constants) but supersymmetry continues to impose a relation between these symmetry violations despite these corrections [7a]. We will see an example of this in Section 29.3.

* * *

The conservation condition (26.7.26) does not uniquely determine either the supercurrent Θ^μ or the associated chiral superfield X . In particular, we may add to Θ^μ a change

$$\Delta\Theta^\mu = \partial^\mu Y, \quad (26.7.53)$$

with Y an arbitrary chiral superfield. The left-hand side of Eq. (26.7.26) is then changed by

$$\gamma_\mu \mathcal{D} \Delta\Theta^\mu = \not{\partial} \mathcal{D} Y.$$

For a left-chiral superfield Y_L the chirality condition $\mathcal{D}_R Y_L = 0$ and the anticommutation relations (26.2.30) give

$$\begin{aligned} \not{\partial} \mathcal{D}_\alpha Y &= -\frac{1}{2} [\{\mathcal{D}_L, \bar{\mathcal{D}}_R\} \mathcal{D}_R]_\alpha Y_L \\ &= -\frac{1}{2} \left[\mathcal{D}_{L\alpha} (\bar{\mathcal{D}}_R \mathcal{D}_R) Y_L + \sum_\beta \bar{\mathcal{D}}_{L\beta} \mathcal{D}_{R\alpha} \mathcal{D}_{L\beta} Y_L \right]. \end{aligned}$$

The matrix $\epsilon\gamma_5$ in $\bar{\mathcal{D}}_{L\beta}$ in the second term on the right of the above expression may be moved to the final operator $\mathcal{D}_{L\beta}$, so that the chirality condition and anticommutation relations give

$$\sum_{\beta} \bar{\mathcal{D}}_{L\beta} \mathcal{D}_{R\alpha} \mathcal{D}_{L\beta} Y_L = - \sum_{\beta} \mathcal{D}_{L\beta} \mathcal{D}_{R\alpha} \bar{\mathcal{D}}_{L\beta} Y_L = 2(\not{\mathcal{D}})_{\alpha} Y_L,$$

and hence

$$\not{\mathcal{D}} Y_L = -\frac{1}{2} [\mathcal{D}(\bar{\mathcal{D}}\mathcal{D})Y_L + 2\not{\mathcal{D}}Y_L] = -\frac{1}{4}\mathcal{D}(\bar{\mathcal{D}}\mathcal{D})Y_L.$$

The same result can be derived in the same way for any right-chiral superfield, and hence it holds also for an arbitrary sum Y of a left-chiral and a right-chiral superfield

$$\gamma_{\mu} \mathcal{D} \Delta \Theta^{\mu} = \not{\mathcal{D}} Y = -\frac{1}{4} \mathcal{D}(\bar{\mathcal{D}}\mathcal{D})Y. \quad (26.7.54)$$

This is of the same form as the conservation condition (26.7.26), with the associated chiral superfield X changed by the chiral superfield

$$\Delta X = -\frac{1}{4}(\bar{\mathcal{D}}\mathcal{D})Y. \quad (26.7.55)$$

It is easy to check that the addition of $\Delta\Theta^{\mu}$ to Θ^{μ} changes $T^{\mu 0}$ and S_{new}^0 only by space derivatives, and therefore does not change the energy-momentum four-vector P^{μ} or the supercharge Q .

We saw in Section 26.6 that any chiral superfield X may be expressed in the form $X = (\bar{\mathcal{D}}\mathcal{D})S$, and hence can be removed by adding a term of the form (26.7.55) to Θ^{μ} with $Y = 4S$. But in general S and the new Θ^{μ} constructed in this way will not be local. This situation is already familiar from our experience with triangle anomalies, discussed in Chapter 22—we saw there that, although it is always possible to construct terms that if added to the Lagrangian density would cancel these anomalies, in general these terms would not be local, and hence must be excluded from the Lagrangian density. There are chiral superfields that can be expressed as $(\bar{\mathcal{D}}\mathcal{D})S$ with S local, and that therefore, if present in the associated chiral superfield X , could be eliminated by adding local terms of the form (26.7.53) to Θ^{μ} . They include, for instance, a term of the form $\text{Re}(k \partial f(\Phi)/\partial \Phi)$, with k an arbitrary complex constant, because the field equations (26.6.15) and (26.6.16) show that $(\bar{\mathcal{D}}\mathcal{D}) \text{Re}(k^* \Phi) = 4 \text{Re}(k \partial f(\Phi)/\partial \Phi)$. But in general the changes in X that can be made in this way are quite limited.

26.8 General Kahler Potentials⁹

There are several circumstances in which it is necessary to consider non-renormalizable Lagrangian densities of the general form (26.3.30)

$$\mathcal{L} = 2 \text{Re}[f(\Phi)]_{\mathcal{F}} + \frac{1}{2}[K(\Phi, \Phi^*)]_D, \quad (26.8.1)$$

⁹This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

where the superpotential f is an arbitrary function of left-chiral scalar superfields Φ_n but not their derivatives, and the Kahler potential K is an arbitrary function of the Φ_n and Φ_n^* but not their derivatives.

This situation arises in effective field theories whose symmetries rule out any renormalizable interactions, or in which the renormalizable interactions all happen to be small. It is often then possible to calculate scattering amplitudes at low energy from tree graphs, using a Lagrangian with the smallest value for some combination of the numbers of derivatives, fermion fields, and any small renormalizable couplings. In Section 19.5 we examined such an effective field theory with no renormalizable couplings, involving nucleons and soft pions. The dynamically broken gauge theories discussed in Section 21.4 provide examples of effective field theories of this sort with small renormalizable couplings. This circumstance also arises in supersymmetric theories whose symmetries do not allow a superpotential, or where the superpotential is for some reason small. We will encounter an example of this sort when we consider the extended $N = 2$ supersymmetric theory of Abelian gauge superfields and gauge-neutral chiral scalar superfields in Section 29.5. We will show there that low-energy scattering amplitudes in this theory are generated by tree graphs using a Lagrangian density of the form (26.8.1), with $f = 0$ and K a function only of the Φ_n and Φ_n^* but no derivatives, plus $\mathcal{F}$ -terms quadratic in gauge superfields. The inclusion of Kahler potentials having arbitrary dependence on Φ_n and Φ_n^* but not on their derivatives is particularly important in effective field theories in which some scalar fields are of the same order of magnitude as the fundamental energy scale of the underlying theory, although all other field values and all energies are much smaller. This will be of interest, for instance, in connection with theories of gravity-mediated supersymmetry breaking, discussed in Section 31.6.

Let us consider how to express the Lagrangian density (26.8.1) in terms of component fields. We did not use the assumption that $f(\Phi)$ is a *cubic* polynomial in deriving Eq. (26.4.4), so this continues to give the $\mathcal{F}$ -terms in the Lagrangian contributed by an arbitrary superpotential. To derive the D -terms, we note that the term in the Kahler

potential of fourth order in θ is

$$\begin{aligned}
K(\Phi, \Phi^*)|_{\theta^4} = & -\frac{1}{8}(\bar{\theta}\gamma_5\theta)^2 \sum_n \left[\frac{\partial K(\phi, \phi^*)}{\partial \phi_n} \square \phi_n + \frac{\partial K(\phi, \phi^*)}{\partial \phi_n^*} \square \phi_n^* \right] \\
& + \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} (\bar{\theta}\gamma_5\theta) [(\bar{\theta}\psi_{mR})(\bar{\theta}\not{\partial}\psi_{nL}) - (\bar{\theta}\psi_{nL})(\bar{\theta}\not{\partial}\psi_{mR})] \\
& + 2 \operatorname{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\theta_L^T \epsilon \psi_{nL})(\theta_L^T \epsilon \psi_{mL})(\theta_L^T \epsilon \theta_L)^* \mathcal{F}_\ell^* \\
& + 2 \operatorname{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\theta}\psi_{mL})(\bar{\theta}\psi_{\ell R})(\bar{\theta}\gamma_5\gamma_\mu\theta) \partial^\mu \phi_n \\
& + \sum_{nmlk} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\theta}\psi_{nL})(\bar{\theta}\psi_{mL})(\bar{\theta}\psi_{\ell R})(\bar{\theta}\psi_{kR}) \\
& - \frac{1}{4} \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} \mathcal{F}_n \mathcal{F}_m^* (\bar{\theta}(1+\gamma_5)\theta) (\bar{\theta}(1-\gamma_5)\theta) \\
& + \frac{1}{4} (\bar{\theta}\gamma_5\gamma^\mu\theta) (\bar{\theta}\gamma_5\gamma^\nu\theta) \sum_{mn} \left[-\frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} \partial_\mu \phi_n \partial_\nu \phi_m^* \right. \\
& \quad \left. + \frac{1}{2} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m} \partial_\mu \phi_n \partial_\nu \phi_m + \frac{1}{2} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n^* \partial \phi_m^*} \partial_\mu \phi_n^* \partial_\nu \phi_m^* \right].
\end{aligned} \tag{26.8.2}$$

We can again use Eqs. (26.A.18) and (26.A.19) as well as Eq. (26.A.9) to put the θ dependence of this expression in the form of an over-all factor $(\bar{\theta}\gamma_5\theta)^2$, and find

$$\begin{aligned}
K(\Phi, \Phi^*)|_{\theta^4} = & \frac{1}{4}(\bar{\theta}\gamma_5\theta)^2 \left\{ -\frac{1}{2} \sum_n \frac{\partial K(\phi, \phi^*)}{\partial \phi_n} \square \phi_n - \frac{1}{2} \sum_n \frac{\partial K(\phi, \phi^*)}{\partial \phi_n^*} \square \phi_n^* \right. \\
& + \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} [(\bar{\psi}_m \not{\partial} \psi_{nL}) + (\bar{\psi}_n \not{\partial} \psi_{mR}) - 2\mathcal{F}_n \mathcal{F}_m^*] \\
& + 2 \operatorname{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_n \psi_{mL}) \mathcal{F}_\ell^* \\
& - 2 \operatorname{Re} \sum_{nml} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_m \gamma^\mu \psi_{\ell R}) \partial_\mu \phi_n \\
& - \frac{1}{2} \sum_{nmlk} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\psi}_n \psi_{mL})(\bar{\psi}_k \psi_{\ell R}) \\
& + \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*} \partial_\mu \phi_n \partial^\mu \phi_m^* \\
& - \frac{1}{2} \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m} \partial_\mu \phi_n \partial^\mu \phi_m \\
& \left. - \frac{1}{2} \sum_{nm} \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n^* \partial \phi_m^*} \partial_\mu \phi_n^* \partial^\mu \phi_m^* \right\}.
\end{aligned} \tag{26.8.3}$$

To make the reality properties of the fermion kinematic terms transparent, we can use

Eq. (26.A.21) to write

$$(\bar{\psi}_n \not{\partial} \psi_{mR}) = (\bar{\psi}_n \not{\partial} \psi_{mL})^*.$$

The D -term of $K(\Phi, \Phi^*)$ is the coefficient of $-(\bar{\theta}\gamma_5\theta)^2/4$ minus half the d'Alembertian of the θ -independent term in $K(\Phi, \Phi^*)$, which is just $K(\phi, \phi^*)$, so

$$\begin{aligned} \frac{1}{2}[K(\Phi, \Phi^*)]_D = & \text{Re} \sum_{nm} \mathcal{G}_{nm} \left[-\frac{1}{2}(\bar{\psi}_m \not{\partial} (1 + \gamma_5) \psi_n) + \mathcal{F}_n \mathcal{F}_m^* - \partial_\mu \phi_n \partial^\mu \phi_m^* \right] \\ & - \text{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_n \psi_{mL}) \mathcal{F}_\ell^* \\ & + \text{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_m \gamma^\mu \psi_{\ell R}) \partial_\mu \phi_n \\ & + \frac{1}{4} \sum_{nm\ell k} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\psi}_n \psi_{mL}) (\bar{\psi}_k \psi_{\ell R}), \end{aligned} \quad (26.8.4)$$

where $\mathcal{G}(\phi, \phi^*)$ is the *Kahler metric*

$$\mathcal{G}_{nm}(\phi, \phi^*) \equiv \frac{\partial^2 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m^*}. \quad (26.8.5)$$

Note that the constant matrix g_{nm} in Eq. (26.4.2) is replaced here with the Kahler metric $\mathcal{G}_{nm}(\phi, \phi^*)$. Because the Kahler metric is field-dependent, we cannot in general make it equal to the unit matrix by a field redefinition, so the total Lagrangian must be left in the form

$$\begin{aligned} \mathcal{L} = & \text{Re} \sum_{nm} \mathcal{G}_{nm} \left[-\frac{1}{2}(\bar{\psi}_m \not{\partial} (1 + \gamma_5) \psi_n) + \mathcal{F}_n \mathcal{F}_m^* - \partial_\mu \phi_n \partial^\mu \phi_m^* \right] \\ & - \text{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_n \psi_{mL}) \mathcal{F}_\ell^* \\ & + \text{Re} \sum_{nm\ell} \frac{\partial^3 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^*} (\bar{\psi}_m \gamma^\mu \psi_{\ell R}) \partial_\mu \phi_n \\ & + \frac{1}{4} \sum_{nm\ell k} \frac{\partial^4 K(\phi, \phi^*)}{\partial \phi_n \partial \phi_m \partial \phi_\ell^* \partial \phi_k^*} (\bar{\psi}_n \psi_{mL}) (\bar{\psi}_k \psi_{\ell R}) \\ & - \text{Re} \sum_{nm} \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} (\bar{\psi}_n \psi_{mL}) + 2 \text{Re} \sum_n \mathcal{F}_n \frac{\partial f(\phi)}{\partial \phi_n}. \end{aligned} \quad (26.8.6)$$

The bilinear $(\bar{\psi}_m \not{\partial} \gamma_5 \psi_n)$ is a total derivative, and could be discarded if $\mathcal{G}_{nm}$ were a constant, but must be kept for general Kahler potentials. This result will be extended to include gauge superfields at the end of Section 27.4.

* * *

As discussed in Section 19.6, the spontaneous breakdown of a global symmetry group G to a subgroup H entails the existence of a set of real massless Goldstone bosons with scalar

fields π_k , for which the term in the Lagrangian with the minimum number of derivatives takes the form

$$\mathcal{L}_{G/H} = - \sum_{k\ell} G_{k\ell}(\pi) \partial_\mu \pi_k \partial^\mu \pi_\ell, \quad (26.8.7)$$

where $G_{k\ell}(\pi)$ is the metric of the coset space G/H . (Theories with Lagrangian densities of this general form are known as *non-linear σ -models*.) By writing the complex fields ϕ_n in terms of their real and imaginary parts, the term $-\sum_{nm} \mathcal{G}_{nm}(\phi, \phi^*) \partial_\mu \phi_n \partial^\mu \phi_m^*$ in the Lagrangian density (26.8.6) can be put in the form (26.8.7), but the converse is not generally true: the condition that a set of real coordinates like the Goldstone boson fields π_k can be interpreted as the real and imaginary parts of a set of complex coordinates like the fields ϕ_n with the metric in these coordinates locally given by Eq. (26.8.5) defines what is called a *Kahler manifold*.¹⁰ But it should not be thought that in the common cases where G/H is *not* a Kahler manifold, it is impossible for G to be spontaneously broken to H while keeping supersymmetry unbroken. What happens in these cases is the appearance of extra massless bosons, which, together with the Goldstone bosons, do form a Kahler manifold.

This is because the superpotential $f(\phi)$ depends on ϕ but not ϕ^* , so if the whole Lagrangian is invariant under a global symmetry group G then the superpotential is automatically invariant under a group $G_{\mathbb{C}}$, the complexification of G : if G consists of transformations $\exp(i \sum_A \theta_A t_A)$ with generators t_A and arbitrary real parameters θ_A , then $G_{\mathbb{C}}$ consists of transformations $\exp(i \sum_A z_A t_A)$ with the same generators and arbitrary complex parameters z_A . (For instance, if G is $U(n)$ then $G_{\mathbb{C}}$ is $GL(n, \mathbb{C})$, the group of all complex non-singular matrices, while if G is $SU(n)$ then $G_{\mathbb{C}}$ is $SL(n, \mathbb{C})$, the group of all complex non-singular matrices with unit determinant.) Likewise, if some stationary point $\phi^{(0)}$ of $f(\phi)$ is left invariant by some subgroup H of G , then it will also be left invariant by a subgroup $H_{\mathbb{C}}$ of $G_{\mathbb{C}}$, the complexification of H . Whether or not G/H is a Kahler manifold, the complexified coset space $G_{\mathbb{C}}/H_{\mathbb{C}}$ is always a Kahler manifold. This follows because $G_{\mathbb{C}}/H_{\mathbb{C}}$ is a complex submanifold of the flat complex space of the ϕ_n , which is a Kahler manifold, and it is a theorem that any complex submanifold of a Kahler manifold is a Kahler manifold [9]. If parameterized by the values of $\phi_n(z) = [\exp(i \sum_A z_A t_A) \phi^{(0)}]_n$, the Kahler manifold $G_{\mathbb{C}}/H_{\mathbb{C}}$ has the metric obtained by embedding it in the flat complex space of the ϕ_n , usually taken with line element $\sum_n d\phi_n d\phi_n^*$.

It is true that $G_{\mathbb{C}}$ is not a symmetry of the whole Lagrangian, but the Goldstone bosons associated with the breakdown of $G_{\mathbb{C}}$ to $H_{\mathbb{C}}$ are nevertheless exactly massless. This

¹⁰The significance of Kahler manifolds in this context was pointed out in an early paper by Zumino [8]. Note that it is not necessary for the metric to be expressible in the form (26.8.5) over the whole manifold with a single Kahler potential $K(\phi, \phi^*)$; it is only necessary that the manifold can be covered in finite overlapping patches in which this holds, with different Kahler potentials in each patch. The simplest example of a Kahler manifold is the flat complex plane, with Kahler potential $|z|^2$. As an example of a coset space G/H that is a Kahler manifold, Zumino gave the case where $G = GL(p, \mathbb{C}) \cdot GL(p+q, \mathbb{C})$ and $H = GL(p, \mathbb{C})$, with p and q arbitrary positive-definite integers, and $GL(N, \mathbb{C})$ the group of complex non-singular $N \cdot N$ matrices. The coset space G/H here has complex coordinates ϕ_n that may be taken as the components of a complex $p \cdot (p+q)$ matrix A , which under G and H undergoes the transformations $A \rightarrow BAC$ and $A \rightarrow BA$, respectively, where B and C are square non-singular complex matrices of dimensionality p and $p+q$, respectively. The Kahler potential in this case is simply $K \propto \ln \det AA^\dagger$.

is guaranteed by the non-renormalization theorem of Section 27.6, or, more simply, by the result of Section 25.4, that massless spin zero particles must come in *pairs* that are related by supersymmetry transformations and that therefore have the same transformation under any global symmetry group G that commutes with supersymmetry.

Appendix Majorana Spinors

This appendix summarizes some algebraic properties of Majorana spinors that are needed in dealing with superfields.

Consider a four-component fermionic Majorana spinor s that like Q or θ may be expressed in the form

$$s = \begin{pmatrix} e\zeta^* \\ \zeta \end{pmatrix}, \quad (26.A.1)$$

where ζ is some two-component spinor and e is the $2 \cdot 2$ matrix

$$e \equiv \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_2.$$

Such a spinor is related to its complex conjugate by

$$s^* = \begin{pmatrix} 0 & e \\ -e & 0 \end{pmatrix} s = -\beta\gamma_5\epsilon s, \quad (26.A.2)$$

where ϵ is the $4 \cdot 4$ matrix

$$\epsilon \equiv \begin{pmatrix} e & 0 \\ 0 & e \end{pmatrix} \quad (26.A.3)$$

and as usual γ_5 and β are $4 \cdot 4$ matrices

$$\gamma_5 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \quad \beta = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix},$$

with $\mathbf{1}$ and 0 understood here as $2 \cdot 2$ submatrices. Taking the transpose of Eq. (26.A.2) and multiplying on the right by β gives the equivalent formula

$$\bar{s} \equiv s^\dagger \beta = s^T \epsilon \gamma_5. \quad (26.A.4)$$

The anticommutation of the spinor components limits the variety of covariants that can be formed from a Majorana spinor. To see this, it will be convenient first to consider the symmetry properties of bilinear covariants, which will be of some interest in themselves. For a pair of Majorana spinors s_1 and s_2 and any $4 \cdot 4$ numerical matrix M , Eq. (26.A.4) gives

$$\begin{aligned} \bar{s}_1 M s_2 &= \sum_{\alpha\beta} s_{1\alpha} s_{2\beta} (\epsilon \gamma_5 M)_{\alpha\beta} = - \sum_{\alpha\beta} s_{2\alpha} s_{1\beta} (\epsilon \gamma_5 M)_{\beta\alpha} \\ &= + \sum_{\alpha\beta} s_{2\alpha} s_{1\beta} (M^T \epsilon \gamma_5)_{\alpha\beta} = \bar{s}_2 (\epsilon \gamma_5)^{-1} M^T \epsilon \gamma_5 s_1, \end{aligned}$$

with the minus sign following the second equal sign arising from the fermionic nature of these spinors. In Section 5.4 we found that the 16 covariant matrices formed from Dirac matrices satisfy

$$M^T = \begin{cases} +\mathcal{C}M\mathcal{C}^{-1}, & M = \mathbf{1}, \gamma_5\gamma_\mu, \gamma_5, \\ -\mathcal{C}M\mathcal{C}^{-1}, & M = \gamma_\mu, [\gamma_\mu, \gamma_\nu], \end{cases} \quad (26.A.5)$$

where $\mathcal{C}$ is the matrix

$$\mathcal{C} = \gamma_2\beta = -\epsilon\gamma_5 = \begin{pmatrix} -e & 0 \\ 0 & e \end{pmatrix}. \quad (26.A.6)$$

It follows that

$$(\bar{s}_1 M s_2) = \begin{cases} +(\bar{s}_2 M s_1), & M = \mathbf{1}, \gamma_5\gamma_\mu, \gamma_5, \\ -(\bar{s}_2 M s_1), & M = \gamma_\mu, [\gamma_\mu, \gamma_\nu]. \end{cases} \quad (26.A.7)$$

In particular, setting $s_1 = s_2 = s$, we find that

$$\bar{s}\gamma_\mu s = \bar{s}[\gamma_\mu, \gamma_\nu]s = 0, \quad (26.A.8)$$

so the only bilinear covariants formed from a single Majorana spinor s are $\bar{s}s$, $\bar{s}\gamma_5\gamma_\mu s$, and $\bar{s}\gamma_5 s$.

In considering the form of the most general superfields, we need to have expressions for the products of two or more Majorana spinors. For two spinors, we recall that any $4 \cdot 4$ matrix may be expanded as a sum of the 16 covariant matrices $\mathbf{1}$, γ_μ , $[\gamma_\mu, \gamma_\nu]$, $\gamma_5\gamma_\mu$, γ_5 . Lorentz invariance tells us that for the matrix $s_\alpha \bar{s}_\beta$ this expansion must take the form

$$\begin{aligned} s\bar{s} &= k_S(\bar{s}s) + k_V\gamma_\mu(\bar{s}\gamma^\mu s) + k_T[\gamma_\mu, \gamma_\nu](\bar{s}[\gamma^\mu, \gamma^\nu]s) \\ &\quad + k_A\gamma_5\gamma_\mu(\bar{s}\gamma_5\gamma^\mu s) + k_P\gamma_5(\bar{s}\gamma_5 s), \end{aligned}$$

where the k 's are constants to be determined. Eq. (26.A.8) shows that we may take $k_V = k_T = 0$. The remaining coefficients may be calculated by multiplying on the right with $\mathbf{1}$, $\gamma_5\gamma^\mu$, and γ_5 and taking the trace, which gives $k_S = -1/4$, $k_A = +1/4$, and $k_P = -1/4$. In this way we find that

$$s\bar{s} = -\frac{1}{4}(\bar{s}s) + \frac{1}{4}\gamma_5\gamma_\mu(\bar{s}\gamma_5\gamma^\mu s) - \frac{1}{4}\gamma_5(\bar{s}\gamma_5 s). \quad (26.A.9)$$

By multiplying on the right with $-\epsilon\gamma_5$ and using Eq. (26.A.4), we may put this in the form

$$s_\alpha s_\beta = \frac{1}{4}(\epsilon\gamma_5)_{\alpha\beta}(\bar{s}s) + \frac{1}{4}(\gamma_\mu\epsilon)_{\alpha\beta}(\bar{s}\gamma_5\gamma^\mu s) + \frac{1}{4}\epsilon_{\alpha\beta}(\bar{s}\gamma_5 s), \quad (26.A.10)$$

or, equivalently,

$$s_\alpha s_\beta = \frac{1}{4}(\epsilon\gamma_5)_{\alpha\beta}(s^T \epsilon\gamma_5 s) + \frac{1}{4}(\gamma_\mu\epsilon)_{\alpha\beta}(s^T \epsilon\gamma^\mu s) + \frac{1}{4}\epsilon_{\alpha\beta}(s^T \epsilon s). \quad (26.A.11)$$

Now consider the product $s_\alpha s_\beta s_\gamma$ of three components of a Majorana spinor s . We can divide s into left- and right-handed parts

$$s = s_L + s_R, \quad s_L = \frac{1}{2}(\mathbf{1} + \gamma_5)s, \quad s_R = \frac{1}{2}(\mathbf{1} - \gamma_5)s. \quad (26.A.12)$$

Each of s_L and s_R has only two independent components, so since the square of any fermionic c-number vanishes, we have $s_{L\alpha}s_{L\beta}s_{L\gamma} = 0$ and $s_{R\alpha}s_{R\beta}s_{R\gamma} = 0$ for all α, β , and γ , and therefore

$$s_\alpha s_\beta s_\gamma = s_{L\alpha}s_{L\beta}s_{R\gamma} + s_{L\alpha}s_{R\beta}s_{L\gamma} + s_{R\alpha}s_{L\beta}s_{L\gamma} + L \leftrightarrow R,$$

with ' $L \leftrightarrow R$ ' denoting the sum of previous terms with labels L and R interchanged. To evaluate this expression, we multiply Eq. (26.A.11) by suitable factors of $(\mathbf{1} + \gamma_5)/2$, and find

$$s_{L\alpha}s_{L\beta} = \frac{1}{4}[\epsilon(\mathbf{1} + \gamma_5)]_{\alpha\beta}(s_L^T \epsilon s_L).$$

If we now multiply this with $s_{R\gamma}$, since $(s_R^T \epsilon s_R)s_{R\gamma} = 0$ we can drop the label L on the spinors in the bilinear $(s_L^T \epsilon s_L)$:

$$s_{L\alpha}s_{L\beta}s_{R\gamma} = \frac{1}{4}[\epsilon(\mathbf{1} + \gamma_5)]_{\alpha\beta}(s^T \epsilon s)s_{R\gamma}.$$

The same arguments also yield

$$s_{R\alpha}s_{R\beta}s_{L\gamma} = \frac{1}{4}[\epsilon(\mathbf{1} - \gamma_5)]_{\alpha\beta}(s^T \epsilon s)s_{L\gamma}.$$

Adding the sum of these two expressions to the same quantity with γ replaced with α or β yields finally

$$\begin{aligned} s_\alpha s_\beta s_\gamma = \frac{1}{4}(s^T \epsilon s) & [\epsilon_{\alpha\beta}s_\gamma - (\epsilon\gamma_5)_{\alpha\beta}(\gamma_5 s)_\gamma - \epsilon_{\alpha\gamma}s_\beta \\ & + (\epsilon\gamma_5)_{\alpha\gamma}(\gamma_5 s)_\beta + \epsilon_{\beta\gamma}s_\alpha - (\epsilon\gamma_5)_{\beta\gamma}(\gamma_5 s)_\alpha]. \end{aligned} \quad (26.A.13)$$

To calculate the products of four Majorana spinor components, we note that $(s^T \epsilon s)$ contains only terms with two s_L s or two s_R s, so

$$(s^T \epsilon s)s_\gamma s_\delta = (s^T \epsilon s)[s_{R\gamma}s_{R\delta} + s_{L\gamma}s_{L\delta}].$$

Using Eq. (26.A.11) to evaluate the sum in square brackets, and noting that

$$(s^T \epsilon s)(s^T \epsilon \gamma_5 s) = (s_L^T \epsilon s_L)(s_R^T \epsilon s_R) - (s_R^T \epsilon s_R)(s_L^T \epsilon s_L) = 0,$$

we find that

$$(s^T \epsilon s)s_\gamma s_\delta = \frac{1}{4}\epsilon_{\gamma\delta}(s^T \epsilon s)^2. \quad (26.A.14)$$

Multiplying Eq. (26.A.13) with s_δ therefore yields the result

$$\begin{aligned} s_\alpha s_\beta s_\gamma s_\delta = \frac{1}{16}(s^T \epsilon s)^2 & [\epsilon_{\alpha\beta}\epsilon_{\gamma\delta} - (\epsilon\gamma_5)_{\alpha\beta}(\epsilon\gamma_5)_{\gamma\delta} - \epsilon_{\alpha\gamma}\epsilon_{\beta\delta} \\ & + (\epsilon\gamma_5)_{\alpha\gamma}(\epsilon\gamma_5)_{\beta\delta} + \epsilon_{\beta\gamma}\epsilon_{\alpha\delta} - (\epsilon\gamma_5)_{\beta\gamma}(\epsilon\gamma_5)_{\alpha\delta}]. \end{aligned} \quad (26.A.15)$$

Any product of five components of s vanishes, so this completes the list of formulas for products of components of a Majorana spinor.

We can use these formulas to derive some additional relations that will be useful in dealing with superfields. By contracting Eq. (26.A.13) with $(\epsilon\gamma_5)_{\beta\gamma}$ and $(\epsilon\gamma_\mu)_{\beta\gamma}$, we find

$$s_\alpha(\bar{s}s) = -(\gamma_5 s)_\alpha(\bar{s}\gamma_5 s) \quad (26.A.16)$$

and

$$s_\alpha(\bar{s}\gamma_5\gamma_\mu s) = -(\gamma_\mu s)_\alpha(\bar{s}\gamma_5 s). \quad (26.A.17)$$

From Eqs. (26.A.16) and (26.A.17) we may derive ‘Fierz’ identities

$$(\bar{s}s)^2 = -(\bar{s}\gamma_5 s)^2, \quad (\bar{s}\gamma_5\gamma_\mu s)(\bar{s}\gamma_5\gamma_\nu s) = -\eta_{\mu\nu}(\bar{s}\gamma_5 s)^2. \quad (26.A.18)$$

Also, Eq. (26.A.14) may be put in a covariant form

$$(\bar{s}\gamma_5 s)s\bar{s} = -\frac{1}{4}\gamma_5(\bar{s}\gamma_5 s)^2. \quad (26.A.19)$$

It will also be useful to record the reality properties of bilinear products of Majorana spinors. For any pair of Majorana spinors s_1, s_2 satisfying the phase convention (26.A.1), Eqs. (26.A.2) and (26.A.4) give

$$(\bar{s}_1 M s_2)^* = -(s_1^\dagger \epsilon \gamma_5 M^* s_2^*) = (\bar{s}_1 \beta \epsilon \gamma_5 M^* \beta \epsilon \gamma_5 s_2).$$

(The minus sign in the middle expression arises from reversing the interchange of s_1 and s_2 that occurs when we take the complex conjugate.) But Eqs. (??) and (26.A.6) give $\beta \epsilon \gamma_5 \gamma_\mu^* \beta \epsilon \gamma_5 = \gamma_\mu$, so

$$\beta \epsilon \gamma_5 M^* \beta \epsilon \gamma_5 = \begin{cases} +M, & M = \mathbf{1}, \gamma_\mu, [\gamma_\mu, \gamma_\mu], \\ -M, & M = \gamma_\mu \gamma_5, \gamma_5, \end{cases} \quad (26.A.20)$$

and therefore

$$(\bar{s}_1 M s_2)^* = \begin{cases} +(\bar{s}_1 M s_2), & M = \mathbf{1}, \gamma_\mu, [\gamma_\mu, \gamma_\mu], \\ -(\bar{s}_1 M s_2), & M = \gamma_\mu \gamma_5, \gamma_5. \end{cases} \quad (26.A.21)$$

Finally, we mention that any spinor u may be written in terms of a pair of Majorana spinors $s_\pm$ as

$$u = s_+ + i s_-, \quad (26.A.22)$$

where

$$s_+ \equiv \frac{1}{2}(u - \beta \epsilon \gamma_5 u^*), \quad s_- \equiv \frac{1}{2i}(u + \beta \epsilon \gamma_5 u^*). \quad (26.A.23)$$

To check that $s_\pm$ are Majorana spinors satisfying Eq. (26.A.2), it is only necessary to recall that $\beta \epsilon \gamma_5$ is real, and that $(\beta \epsilon \gamma_5)^2 = \mathbf{1}$.

Problems

1. Using the direct technique of Section 26.1 in the case of $N = 2$ supersymmetry, find the supersymmetry transformation rules of the massive field supermultiplet with just one Majorana spinor field and two complex scalars.
2. Calculate the component fields of a time-reversed superfield

$$T^{-1}S(x, \theta)T$$

in terms of the components of the superfield $S(x, \theta)$. What sort of superfield do we get by the time reversal of a left-chiral superfield? Of a linear superfield?

3. Consider the $N = 1$ supersymmetric theory of a single left-chiral superfield Φ . In superfield notation, list all the terms of dimensionality 5 involving Φ and/or Φ^* that might be added to the Lagrangian density.
4. Consider the theory of three left-chiral scalar superfields Φ_1 , Φ_2 , and Φ_3 , with a conventional kinematic term, and with a superpotential

$$f(\Phi_1, \Phi_2, \Phi_3) = \Phi_1 \Phi_3^2 + \Phi_2 (\Phi_3^2 + a),$$

where a is a non-zero real constant. Show that this is a theory with spontaneously broken supersymmetry. Find the minimum value of the potential. Express the field of the goldstino in terms of the fermionic components of Φ_1 , Φ_2 , and Φ_3 .

5. Find all the components of the current superfield for the action (26.6.9), in terms of the components of the left-chiral superfields Φ_n and derivatives of the superpotential f and the Kahler potential K .
6. Check that the supersymmetry current given by Eqs. (26.7.18) and (26.7.10) is related to the ω -component of the superfield (26.7.21) by Eq. (26.7.20).

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